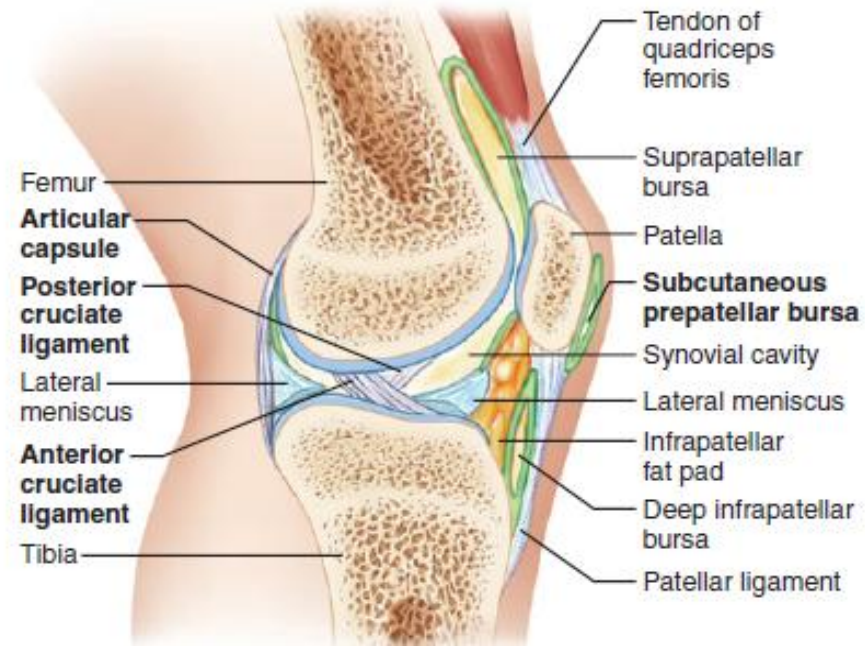
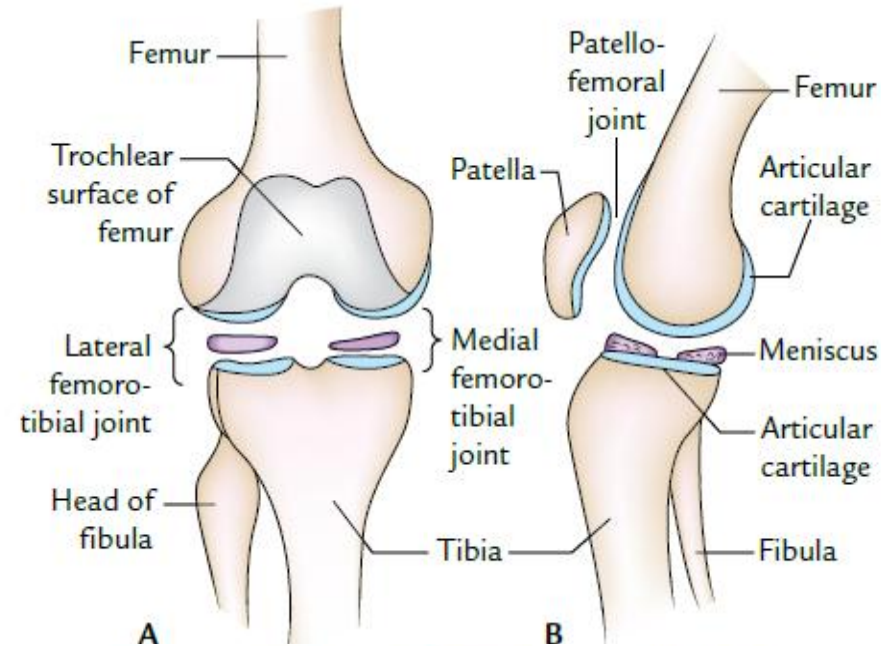


THE KNEE JOINT

Knee Joint & Stabilizing Ligaments

- The knee joint is the **largest and most complicated joint in the body.**
- It is the major weight-bearing joint in the body.
- It is prone to undergo degenerative changes with advanced age
- There are two ligaments in the knee joint
- **Extra capsular ligament**
- **Intracapsular ligament**



Knee Joint & Stabilizing Ligaments

❑ Extra capsular Ligaments

■ Ligamentum Patellae

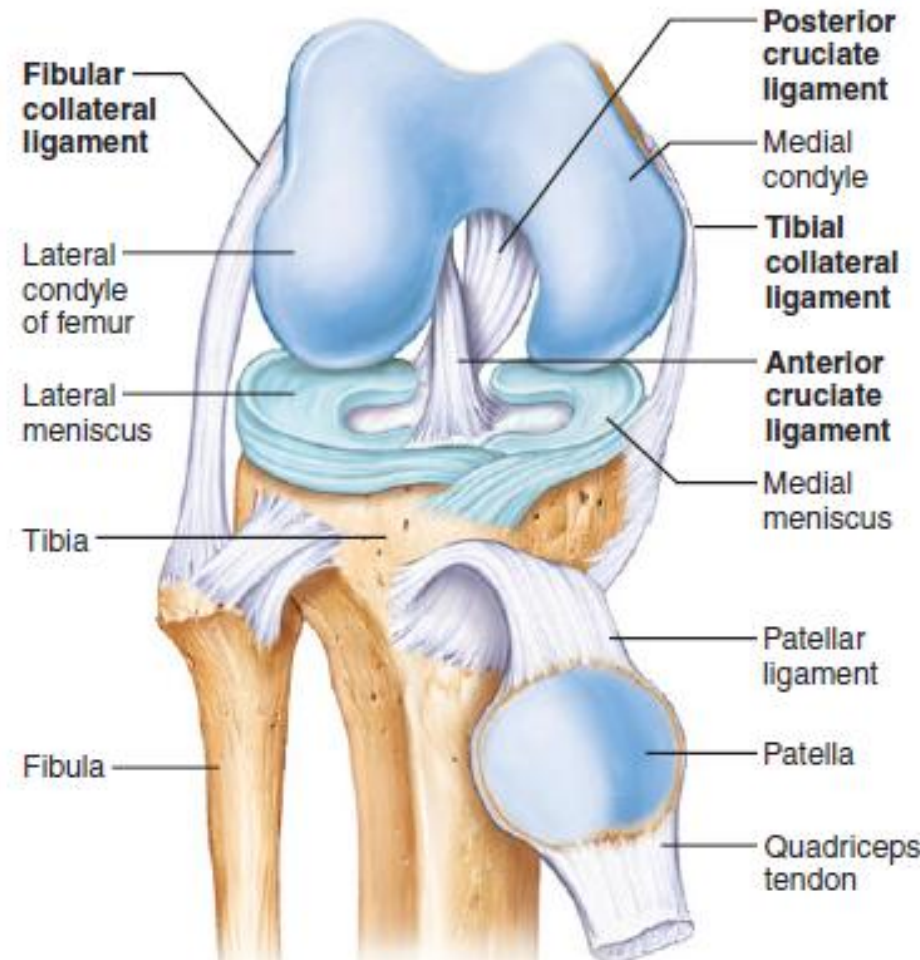
(a continuation of the Quadriceps Femoris muscle)

■ Lateral Collateral Lig.= FCL

■ Medial Collateral Lig.= TCL

■ Oblique Popliteal Lig.

(derived from the Semimembranosus muscle).



Knee Joint & Stabilizing Ligaments

□ Intracapsular Ligaments

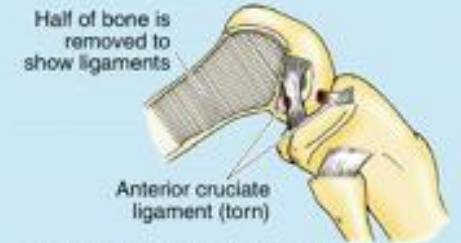
- **Cruciate Ligaments** : 2 strong ligaments that cross each other within the joint cavity .
 - ACL
 - PCL
- **The two menisci (medial and lateral)**



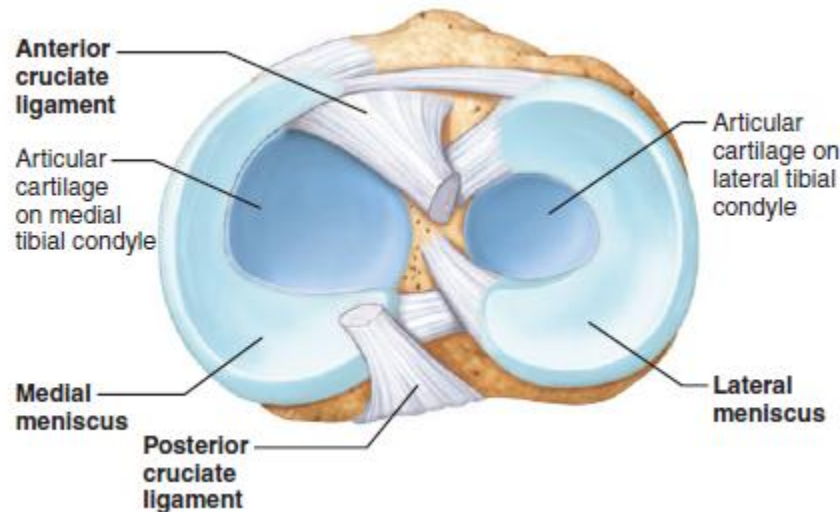
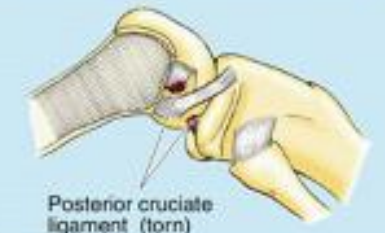
B Anterior drawer sign (ACL)



C Post



The anterior cruciate ligament (ACL) prevents the femur from sliding posteriorly on the tibia and hyperextension of the knee and limits medial rotation of the femur when the foot is on the ground and the leg is flexed.



Medial meniscus

- C-shaped/semilunar in shape
- Attached to the medial collateral ligament
- More prone to injury

Lateral meniscus

- "O"-shaped/circular in shape
- Attached to the tendon popliteus muscle
- Less prone to injury



Anterior Drawer Test



Posterior Drawer Test

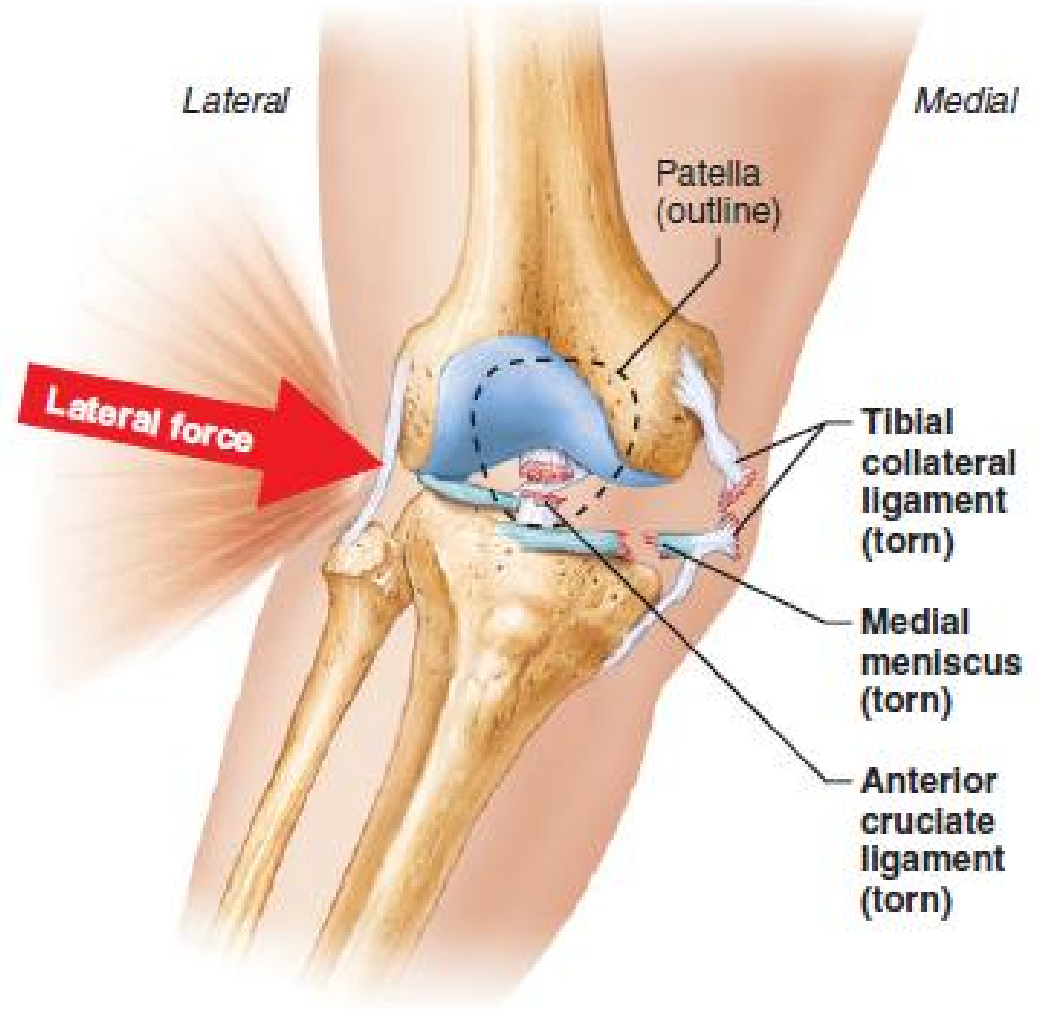


Knee Joint Injury

- Injury of three structures at the same time

☐ Unhappy triad injury

- ACL
- PCL
- MM



A close-up, slightly blurred background image of a doctor in a white lab coat. A blue stethoscope is draped around the neck. On the right side, a circular device with the text 'Spirit ORIGINAL' and 'CE' is visible. Below it, a white pen with 'LAMAL' written on it and a pair of orange-handled scissors are partially visible. The text is overlaid in a white, italicized serif font.

*A good Physician treats the
disease, the great Physician treats
the patient who has the disease.*

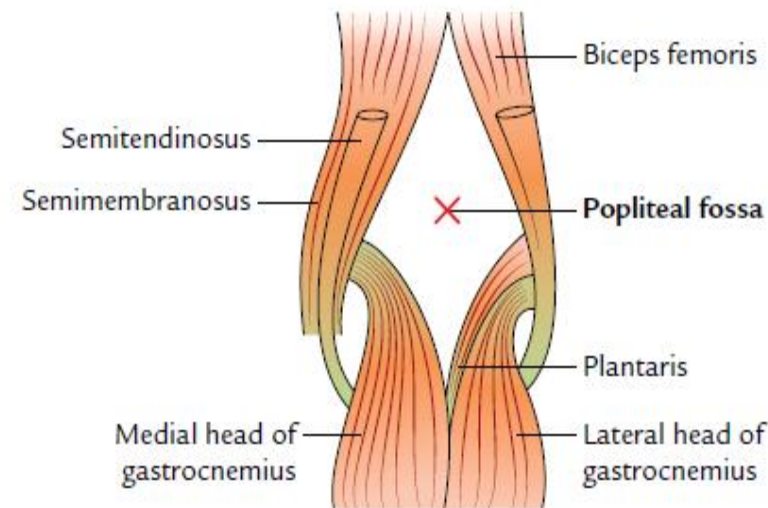
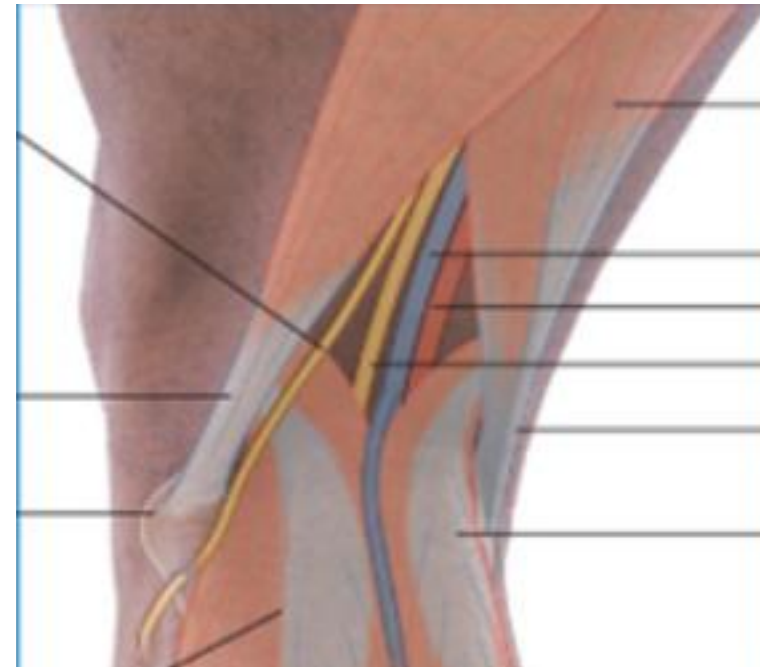
THE POPLITEAL FOSSA

Popliteal fossa

- It is a **fat-filled diamond-shaped space posterior to the knee**
- All important vessels and nerves from the thigh to the leg pass through it
- Distal continuation of the adductor canal

□ Boundaries

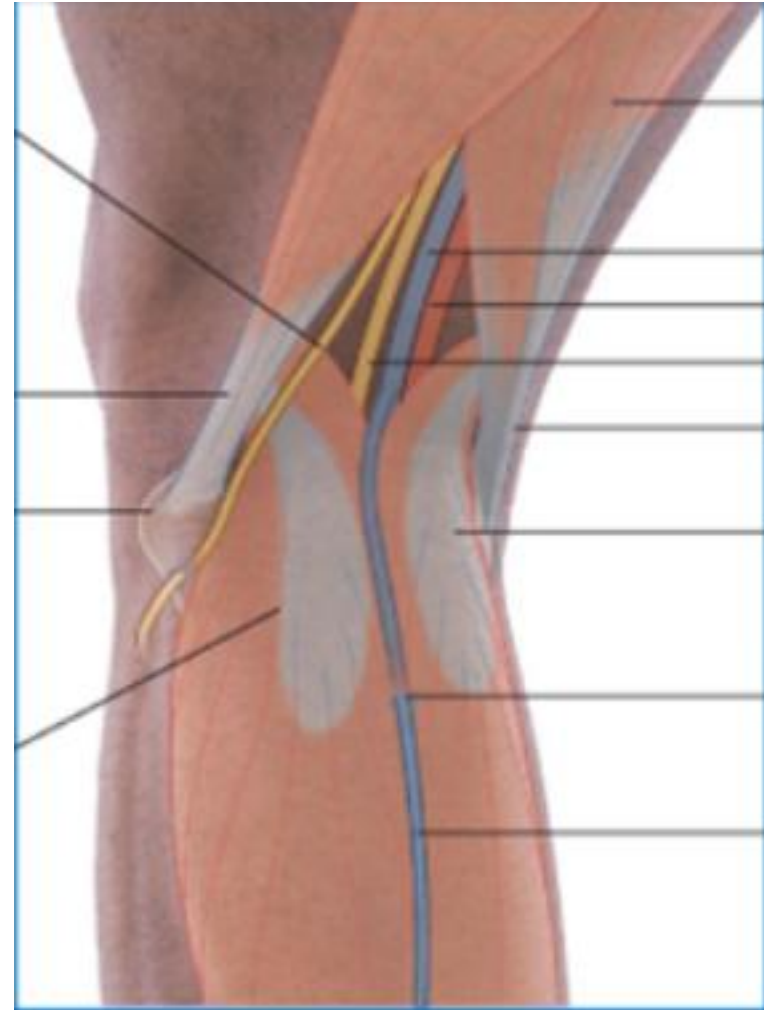
- **Laterally & inferiorly** – lateral head of gastrocnemius
- **Medially & inferiorly**- medial head of gastrocnemius
- **Laterally & superiorly** - biceps femoris
- **Medially & superiorly** – semimembranosus



Popliteal fossa

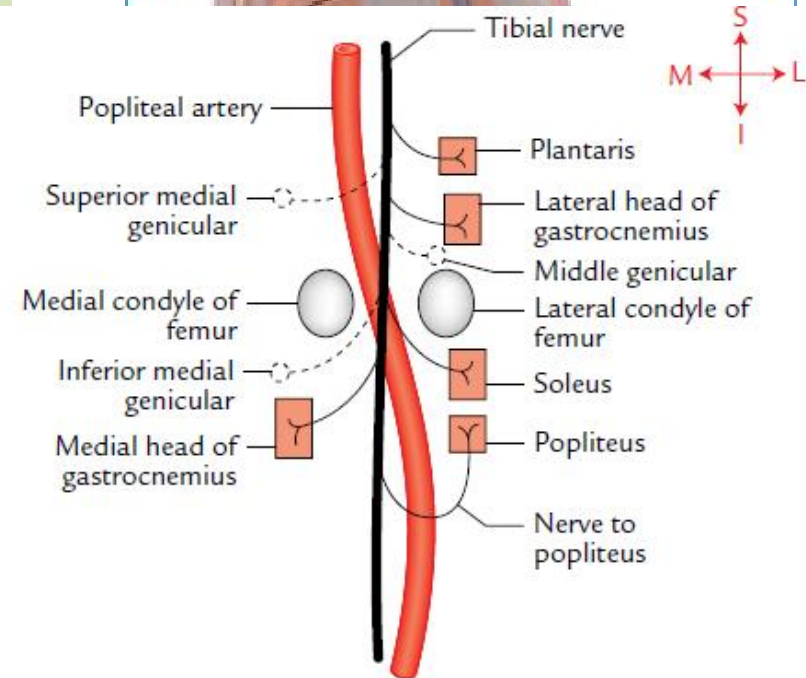
□ Contents

- Popliteal vessels (artery, vein and lymphatics)
- Tibial and common peroneal (fibular) nerves
- Small saphenous vein
- Popliteal lymph nodes

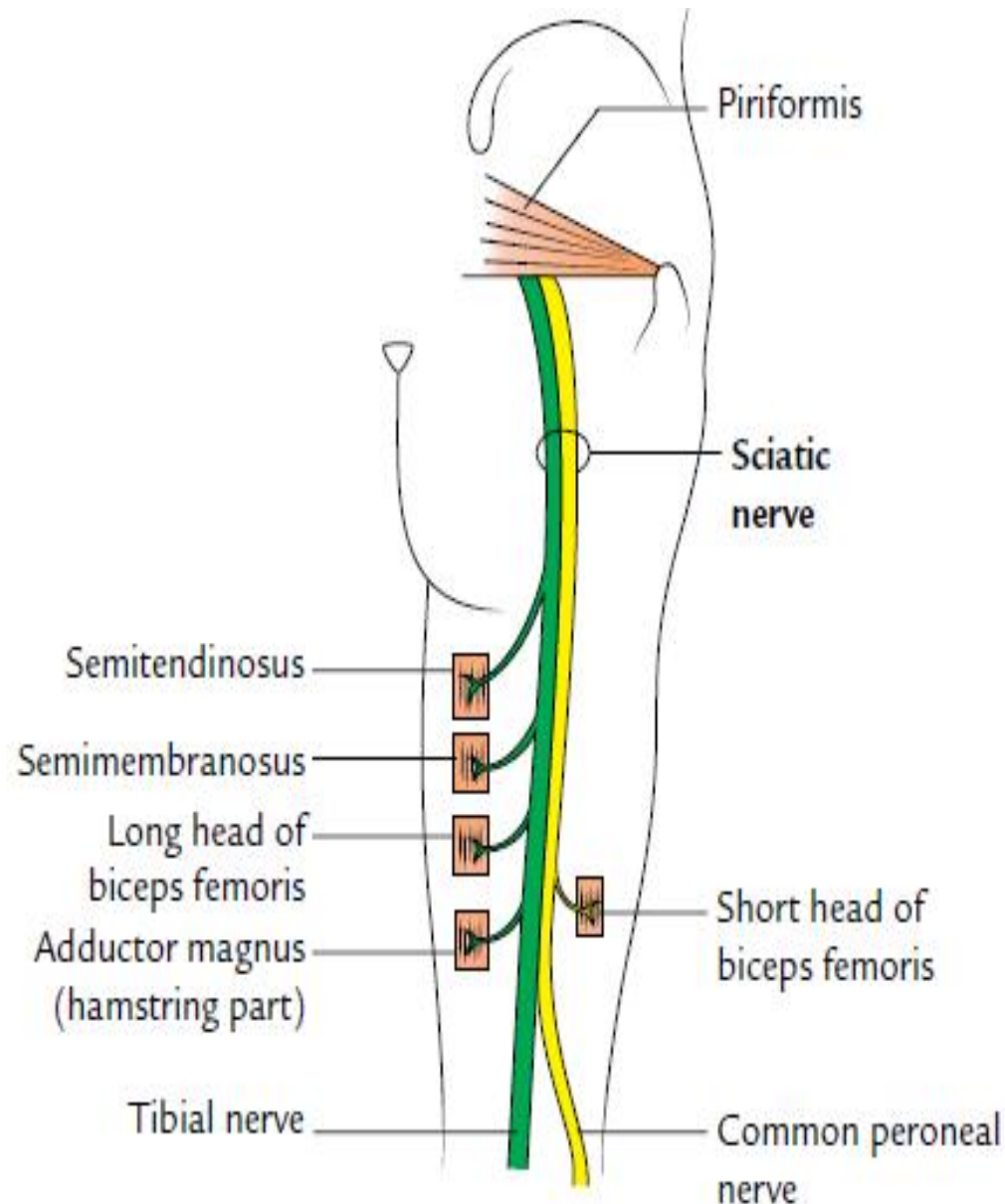


Nerves in popliteal fossa

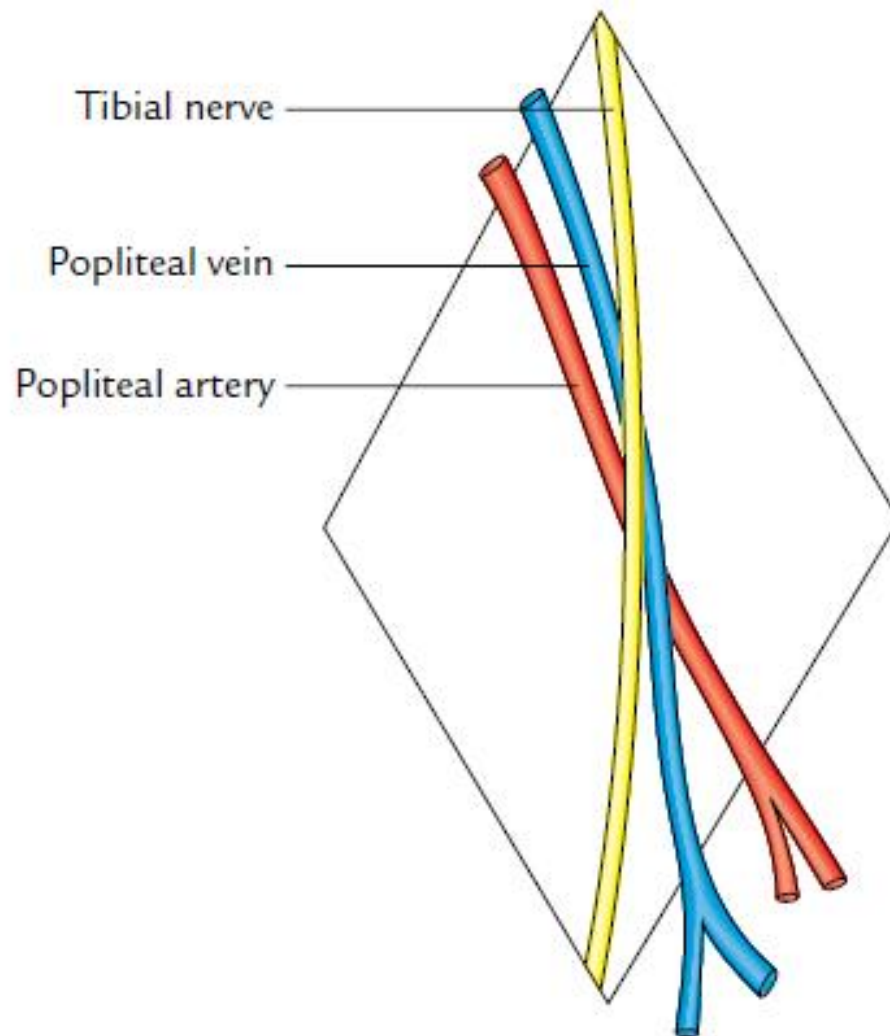
- The sciatic nerve usually ends at the **superior angle** of the **popliteal fossa** by dividing into tibial and common fibular nerves
- **Tibial nerve**
 - The medial, larger terminal branch of the sciatic nerve
 - The **most superficial of the three main central components** of the popliteal fossa (**nerve, vein, and artery**)



Nerves in the Popliteal Fossa



Relations of NV structures in the popliteal fossa



In the upper part
(from lateral to medial)



In the middle part
(from superior to deep)

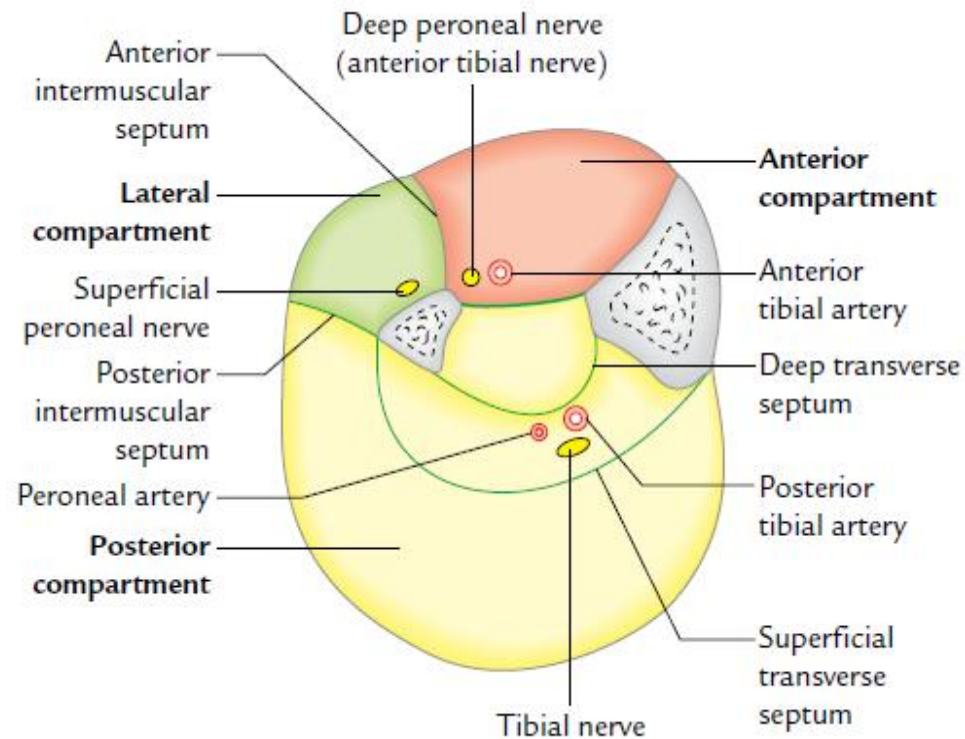


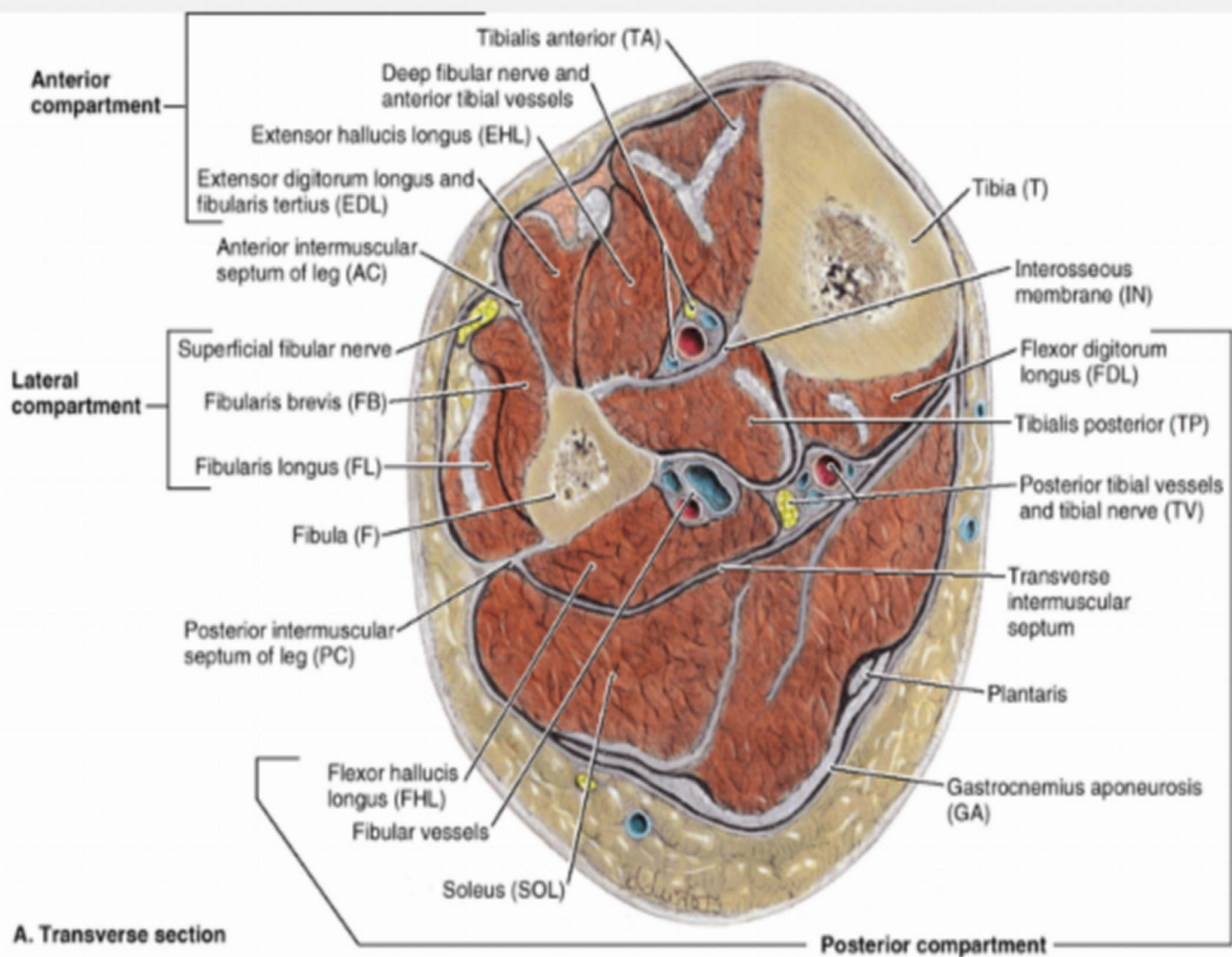
In the lower part
(from lateral to medial)



Anatomy of the LEG

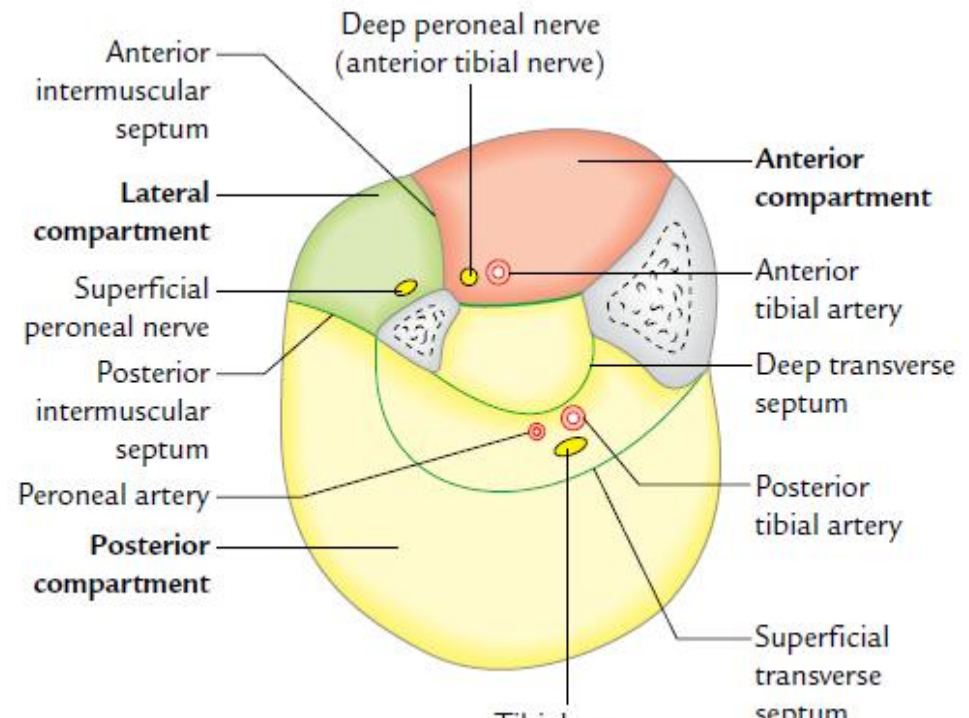
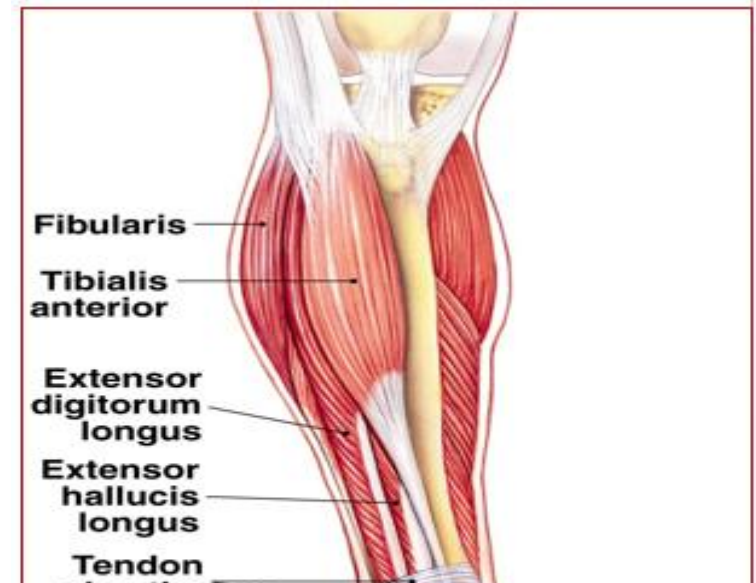
- The leg (crus) is the portion of the lower limb between the knee and the ankle
- It is divided into 3 compartments by anterior and posterior intermuscular septa and interosseous membrane
 - Anterior crural compartment
 - Posterior crural compartment
 - Lateral (fibular or peroneal) compartment





Anterior compartment of LEG

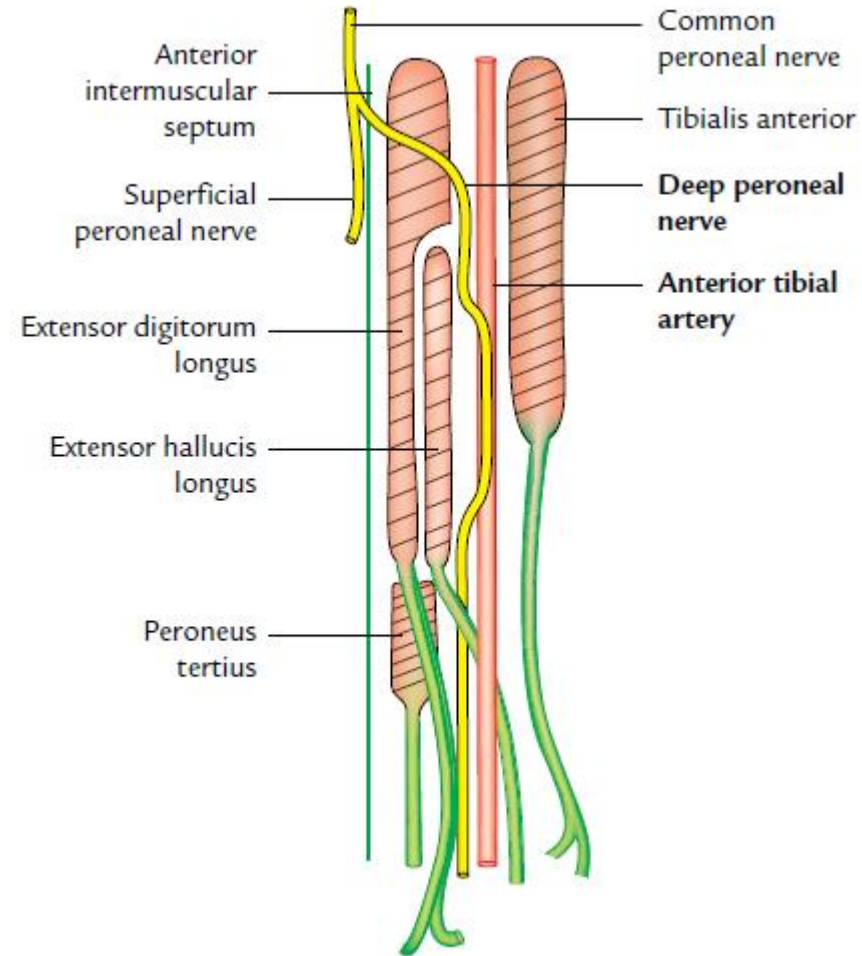
- Dorsi flexor (extensor) compartment
- Located anterior to the interosseous membrane
- **It contain four muscles**
- **Anterior compartment is innervated by deep fibular nerve**
- **All supply by anterior tibial artery**



Anterior compartment of LEG

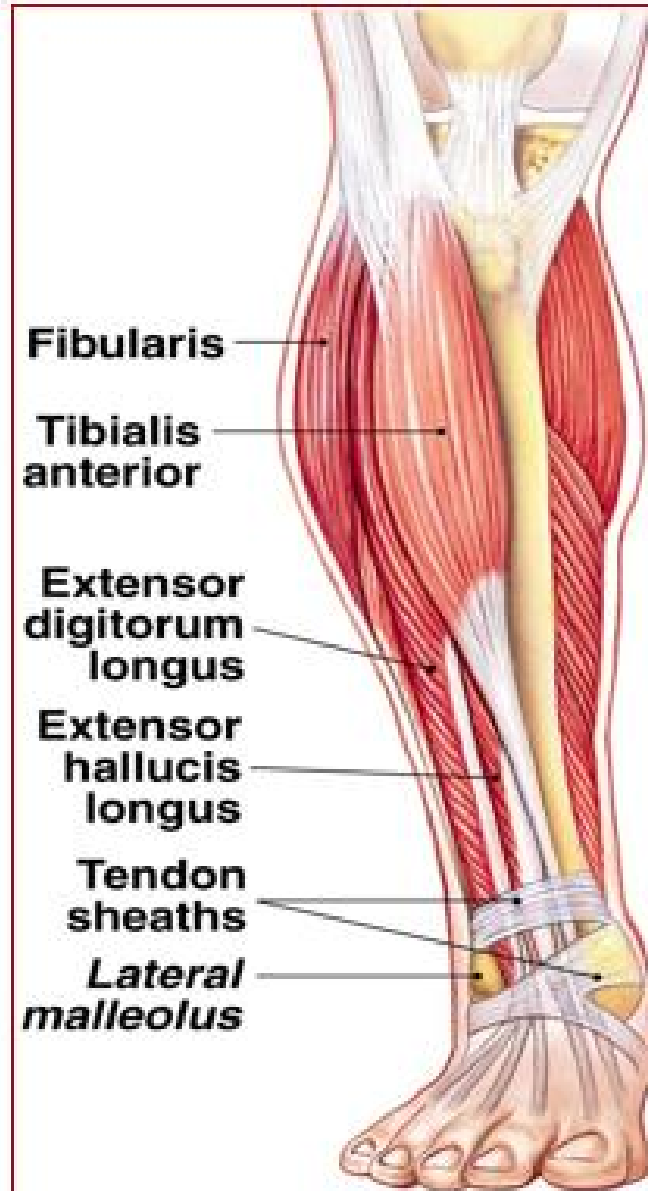
- **Deep fibular nerve**

- One of the two terminal branches of the common fibular nerve
- The nerve of the anterior compartment
- Arises between the fibularis longus muscle and the neck of the fibula
- **After entering the compartment, the nerve accompanies the anterior tibial artery**



Muscles of anterior compartment of LEG

- Primarily **toes extensors** and **dorsiflexors** of ankle joint
 - **Tibialis anterior (inversion)**
 - **Extensor digitorum longus**
 - **Extensor hallucis longus**
 - **Peroneus tertius (eversion)**
- **Nerve supply: deep fibular nv.**
- Arterial supply by **anterior tibial artery**

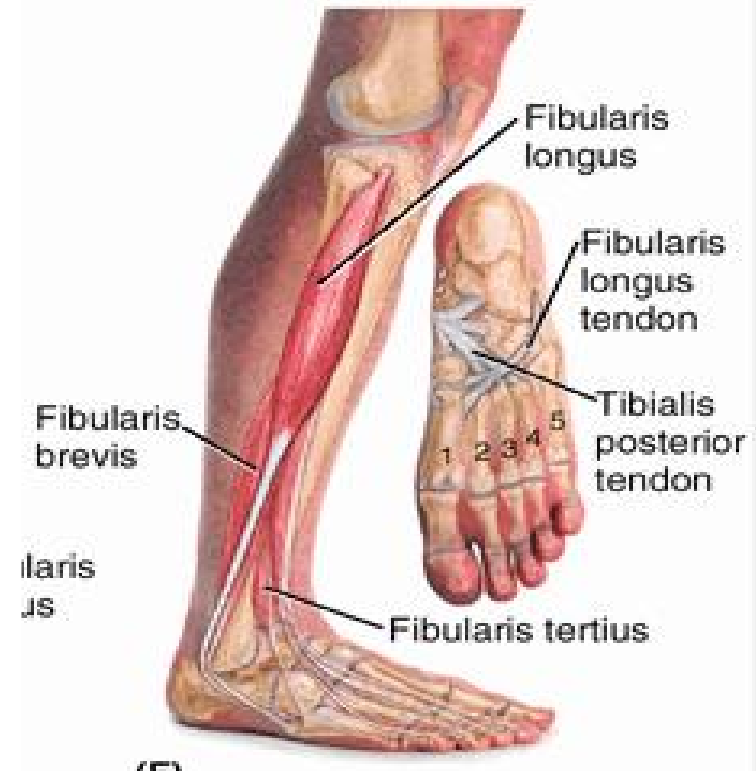


Lateral compartment of LEG

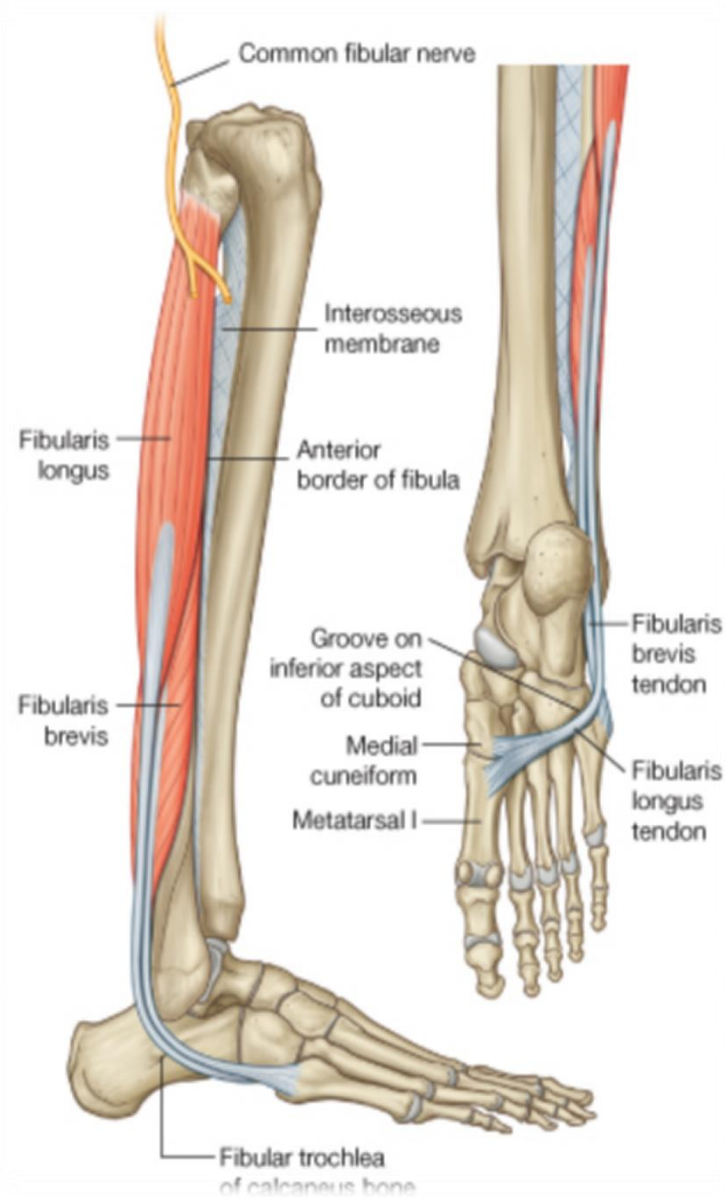
Lateral compartment

- There are only two muscles in the lateral compartment
 - **Fibularis longus**
 - Its name is due to long tendon and superficial to the brevis
 - **Fibularis brevis**
 - It is because it has short tendon, deep to fibularis longus

NB : the lateral compartment of leg muscle are innervated by **superficial fibular nerve** terminal branch of the common fibular nerve



Lateral compartment of LEG



Muscles of posterior compartment of LEG

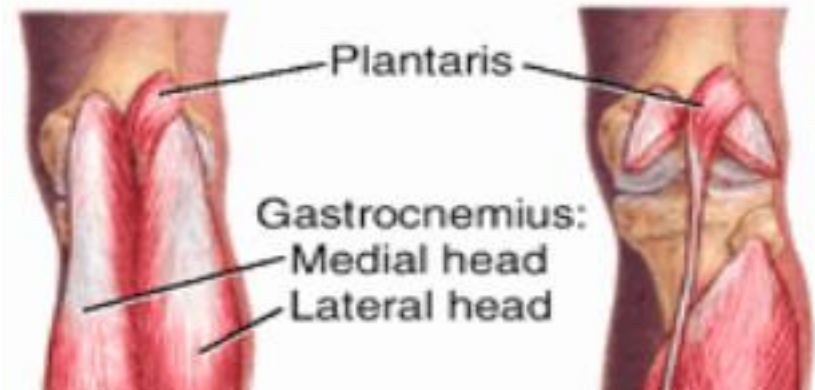
- Action : Plantar flexors and flex the toes

SUPERFICIAL GROUP

- Has 3 muscles:

- Gastro (heart)
- Soleus
- Plantar

- These three muscles are called the **triceps surae**



Clinical correlation

Calf muscle pump and peripheral heart: The gastrocnemius and soleus together make a '*calf muscle pump*,' which facilitates the venous return from the lower limb.

The soleus muscle is regarded as the *peripheral heart* because it houses large venous sinuses called soleal sinuses which communicate with the superficial veins by perforating veins and with deep veins directly.

Thus contraction of soleus helps in sucking the blood from superficial veins and propelling it into deep veins.

Gastrocnemius strain

- Gastrocnemius strain (**tennis leg**) is a painful calf injury resulting from partial tearing of the **medial belly of the gastrocnemius** at or near its musculotendinous junction
- It is caused by the muscle during plantar flexion



Clinical correlation

Tendon grafting: The plantaris is a vestigial muscle in human beings and is absent in 5–10% of people. It has insignificant role either as flexor of the knee or as plantar flexor of the ankle; hence, the tendon of plantaris is used for grafting (e.g., reconstructive surgery of the tendons of the hand).

Muscles of posterior compartment of LEG

- **DEEP GROUP OF MUSCLES CONTAINS**

- **Popliteus**

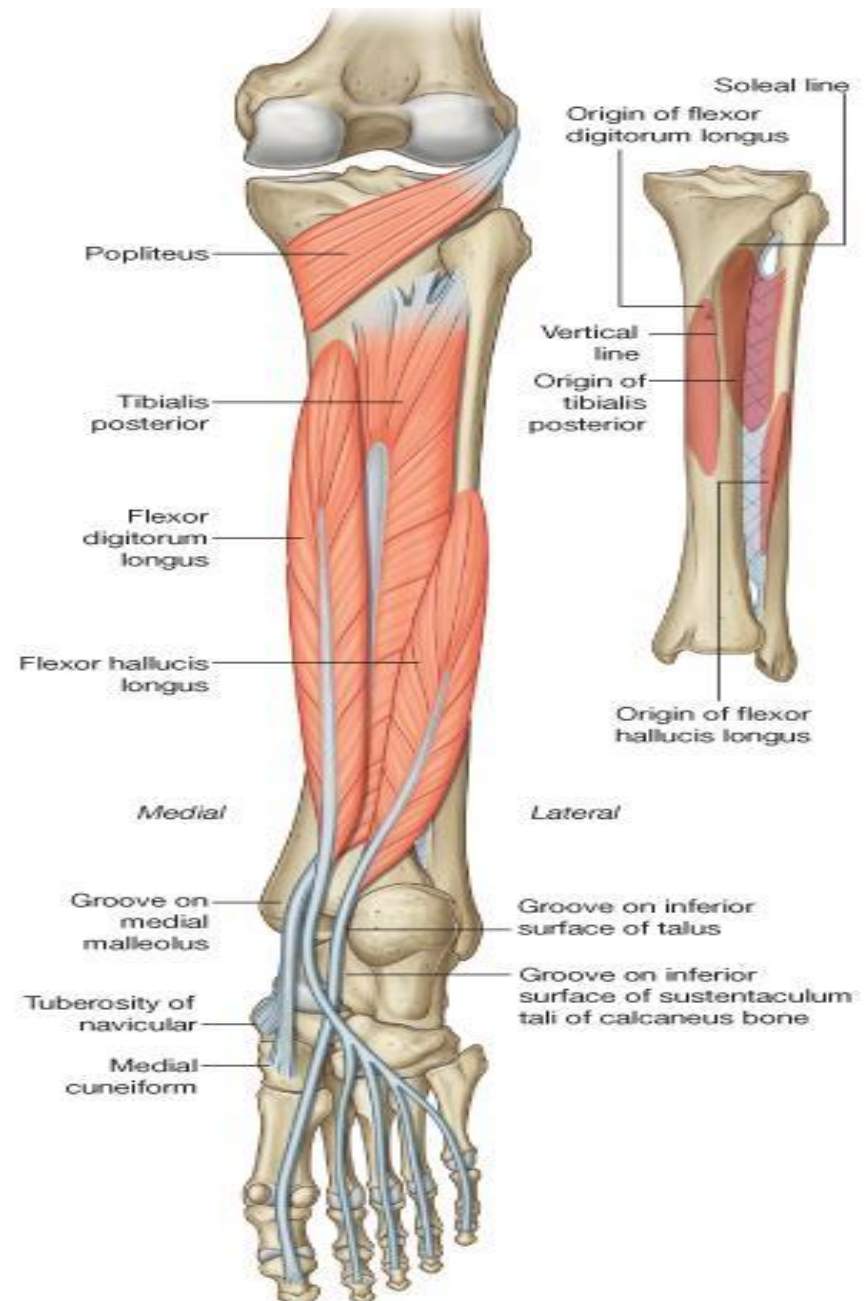
(unlock the knee joint)

- **Flexor hallucis longus**

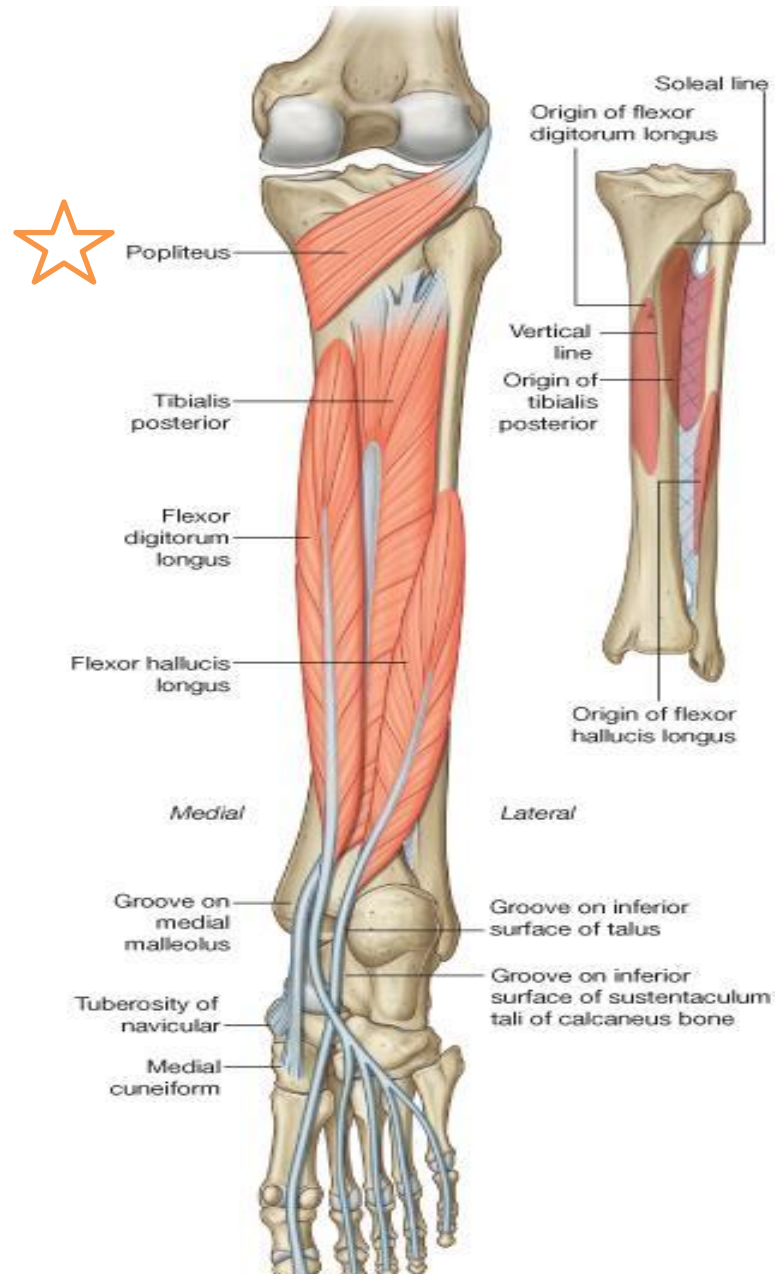
Invert great toe

- **Flexor digitorum longus**

- **Tibialis posterior**

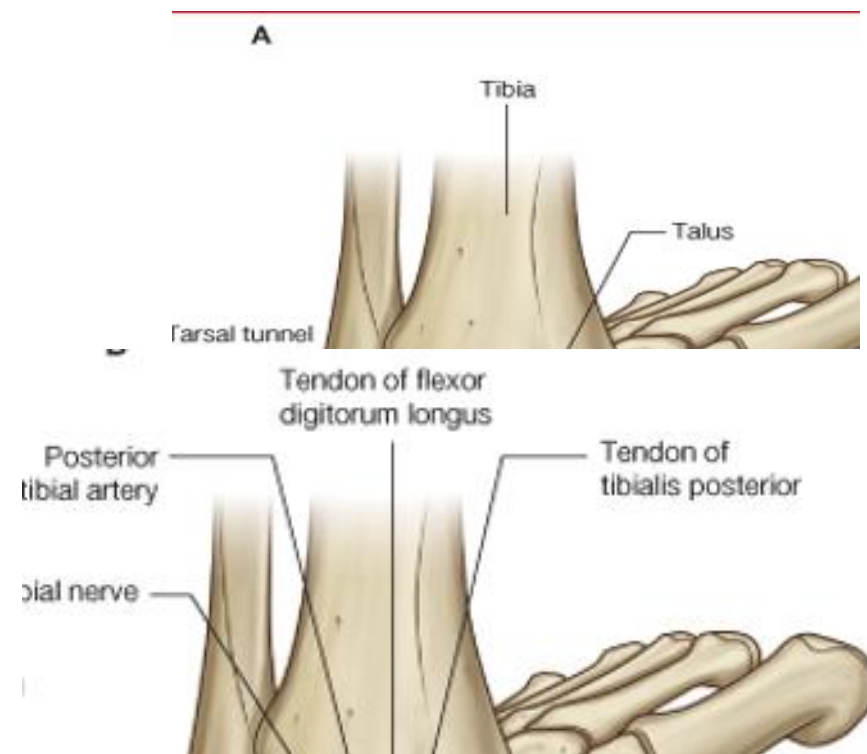


Popliteus



Tarsal tunnel

- Medial malleolus to calcaneus
- Content – from medial to lateral
 - Tendon of tibialis posterior
 - Tendons of flexor digitorum longus
 - Termination of posterior tibial artery and beginning of lateral and medial plantar arteries
 - Termination of nerve and beginning of medial and lateral plantar nerves
 - Tendon of flexor digitorum longus

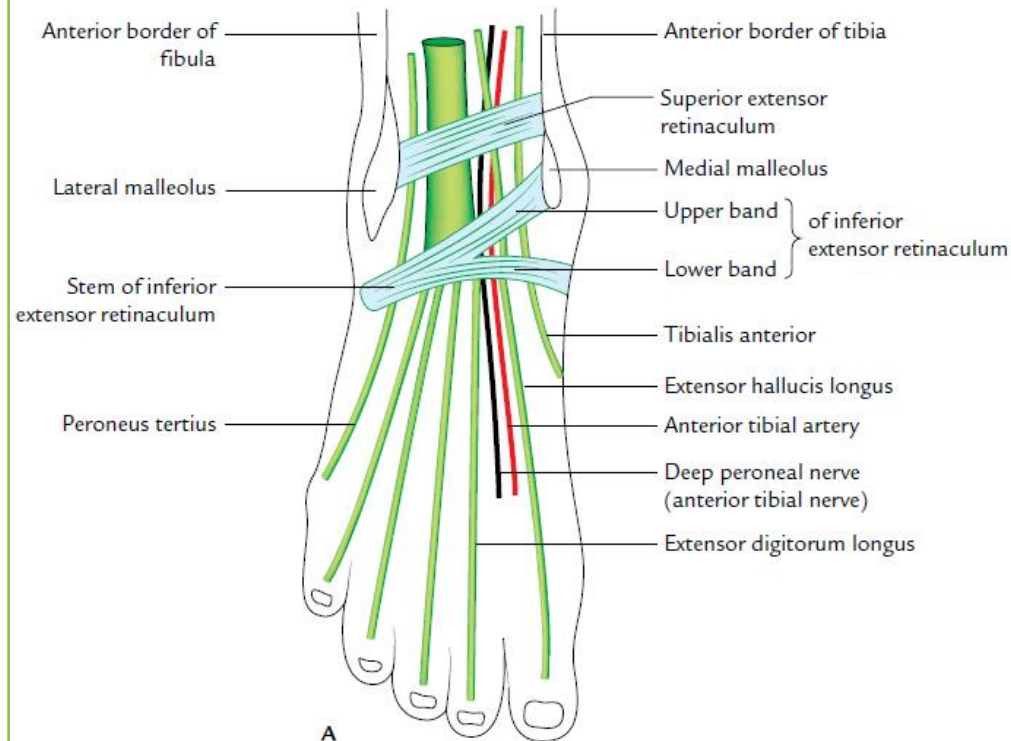


Clinical correlation

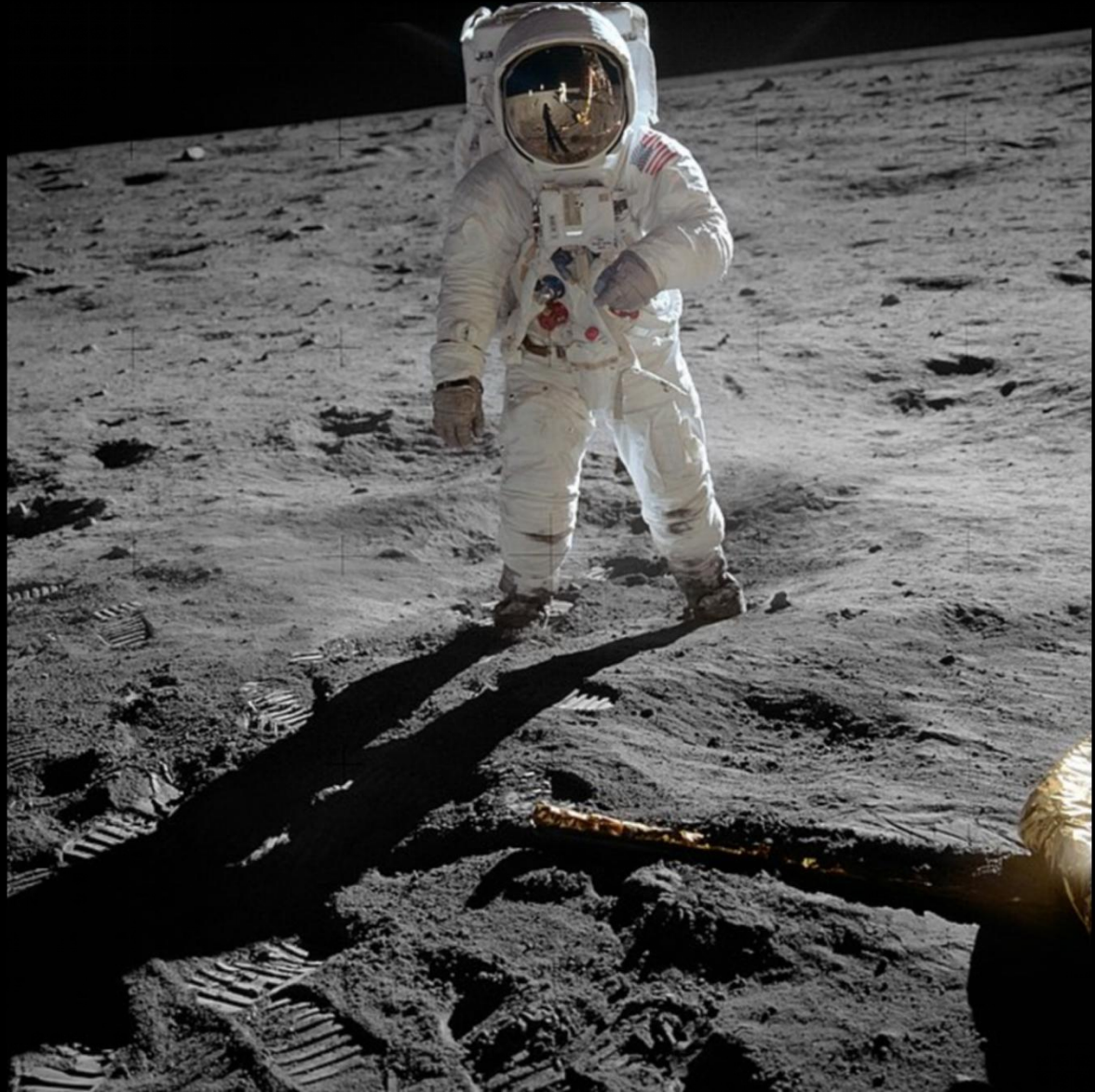
Tarsal tunnel syndrome: If the tibial nerve is compressed deep to the flexor retinaculum (in the osseofibrous tunnel behind the medial malleolus) it leads to a clinical condition called *tarsal tunnel syndrome*. Clinically, it presents as burning, tingling, and pain in the sole of the foot. These symptoms are relieved by dividing the flexor retinaculum surgically.

Extensor retinaculum

- Has **superior and inferior portions**
- **Prevents the tendons** of peroneus tertius, extensor hallucis longus and extensor digitorum longus from **bowstringing**.
- There is also a superior and an inferior peroneal (fibular) retinaculum for the tendons of the peroneus brevis and longus.



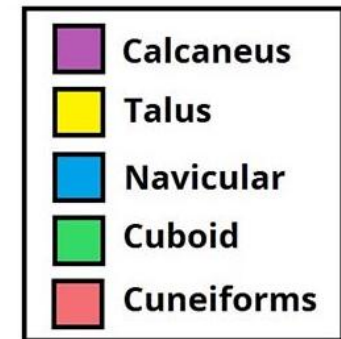
That's one small step for man, one giant leap for mankind!



THE FOOT

Anatomy of Foot

- **Foot is a region of lower limb distal to ankle**
- Provides platform for supporting the weight of the body
- The foot and its bones may be considered in terms of three anatomical and functional parts:
 - **The hindfoot:** talus and calcaneus
 - **The midfoot:** navicular, cuboid, and cuneiforms
 - **The forefoot:** metatarsals and phalanges



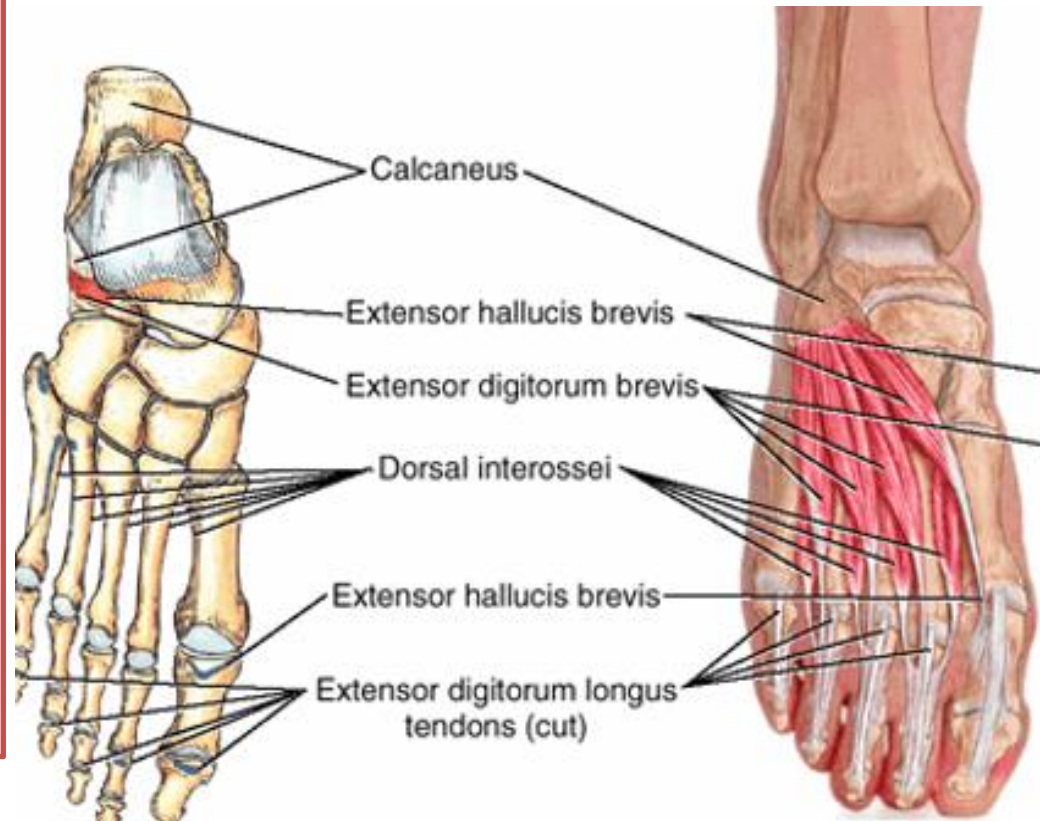
Regions of the foot

- The plantar region (sole): the part contacting the ground.
- The dorsal region (dorsum of the foot): the part directed superiorly.
- The heel region (heel): the sole underlying the calcaneus.
- The ball of the foot: the sole underlying the heads of the medial two metatarsals.



Dorsum of the foot

- **Two Muscles**
 - **Extensor digitorum brevis**
 - **Extensor hallucis brevis**
(medial slip of the extensor digitorum brevis)
- **Nerve supply – deep fibular nerve**



Plantar muscles

- There are 4 layers of plantar muscles, organized as:
- **“3232”**
 - First layer: 3 muscles
 - Second layer: 2 muscles and 2 tendons
 - Third layer: 3 muscles
 - Fourth layer: 2 muscles and 2 tendons

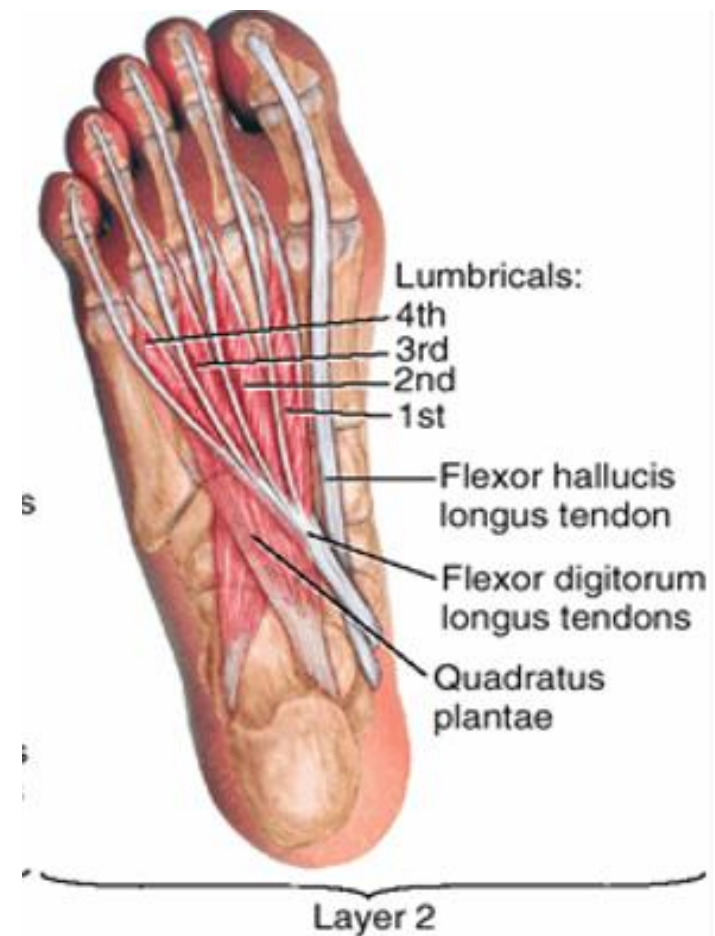
First layer

- **Formed by 3 muscles which arise from the calcaneus**
- Consists of a flexor between 2 abductors
- **Abductor hallucis**
 - innervated by the **medial plantar nerve**
- **Abductor digiti minimi**
 - is innervated by the **lateral plantar nerve**
- **Flexor digitorum brevis**
 - is innervated by the **medial plantar nerve**



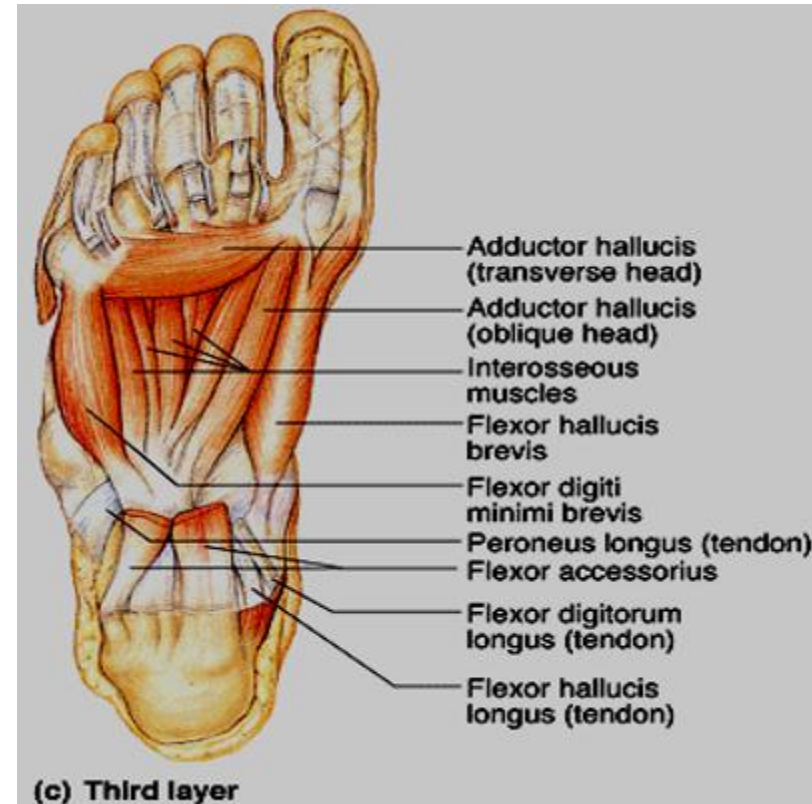
Second layer

- Muscles of the second layer flex, adduct, and abduct the toes
- Formed by 2 muscles and 2 tendons
 - **Flexor digitorum longus tendon**
 - **Flexor hallucis longus tendon**
 - **Quadratus plantae** (flexor digitorum accessorius)
 - It is innervated by the **lateral plantar nerve**
 - **Lumbricals**
 - lumbrical 1 is innervated by the **medial plantar nerve**; lumbricals 2, 3 and 4 are innervated by the **lateral plantar nerve**



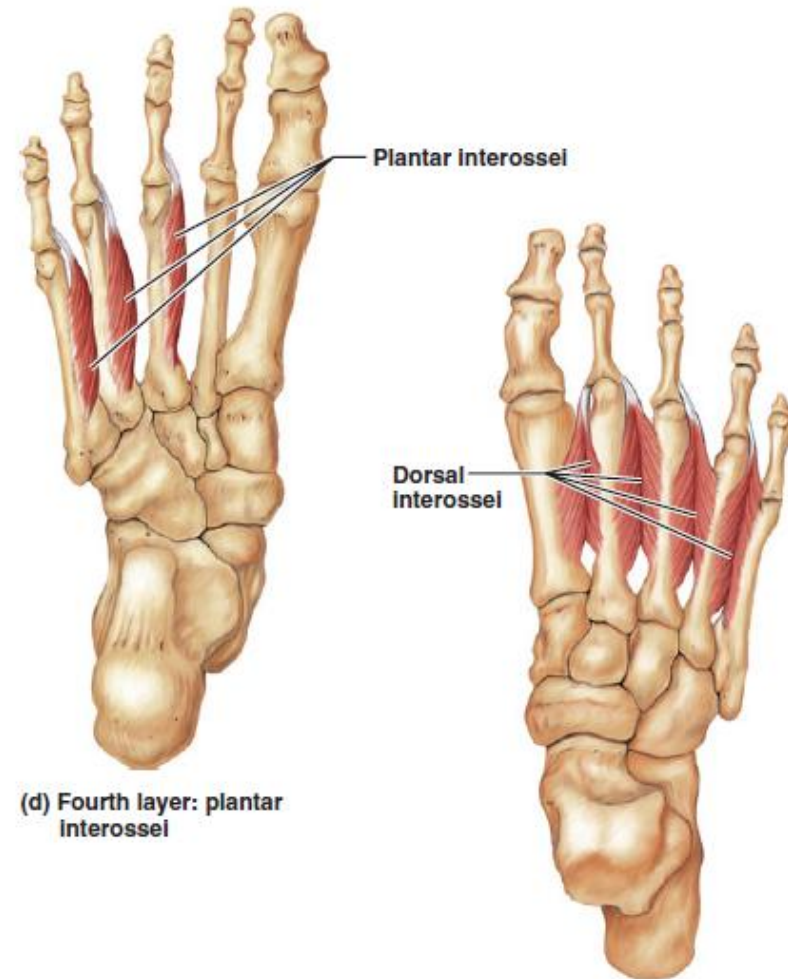
Third layer

- Formed by 3 muscles:
- **Adductor hallucis**
 - Origin
 - Oblique head: from the tarsal bones
 - Transverse head: from the metatarsal heads
 - Insertion: lateral base of proximal phalanx of the hallux
- Innervation: *lateral plantar nerve*
- **Flexor hallucis brevis**
 - Innervation: medial plantar nerve
- **Flexor digiti minimi**
 - Innervation: lateral plantar nerve



Fourth layer

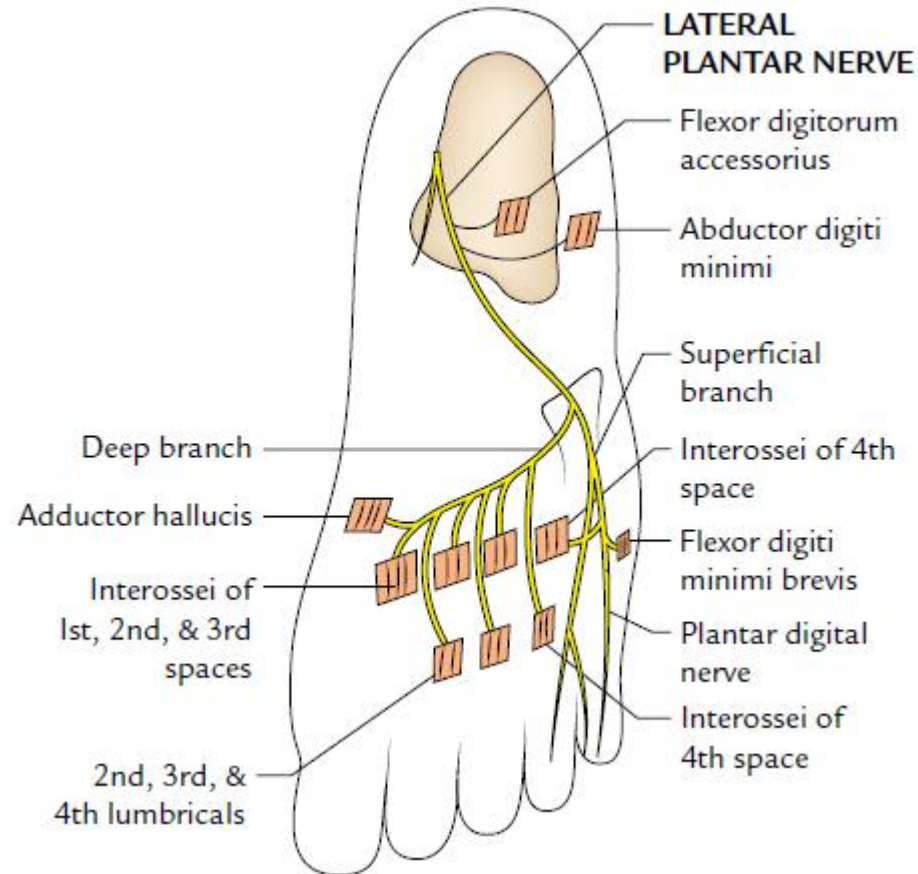
- Formed by 2 muscles and 2 tendons
- **7 interossei (4 dorsal and 3 plantar)**
 - keep the foot from spreading by holding the heads of the metatarsals together
 - also flex the metatarsophalangeal joints and extend the interphalangeal joints
 - are innervated by the *lateral plantar nerve*
 - Abduction and adduction are in reference to toe #2.
- **Tibialis posterior tendon**
- **Peroneus (fibularis) longus tendon**



Nerves of foot

❑ Lateral plantar nerve

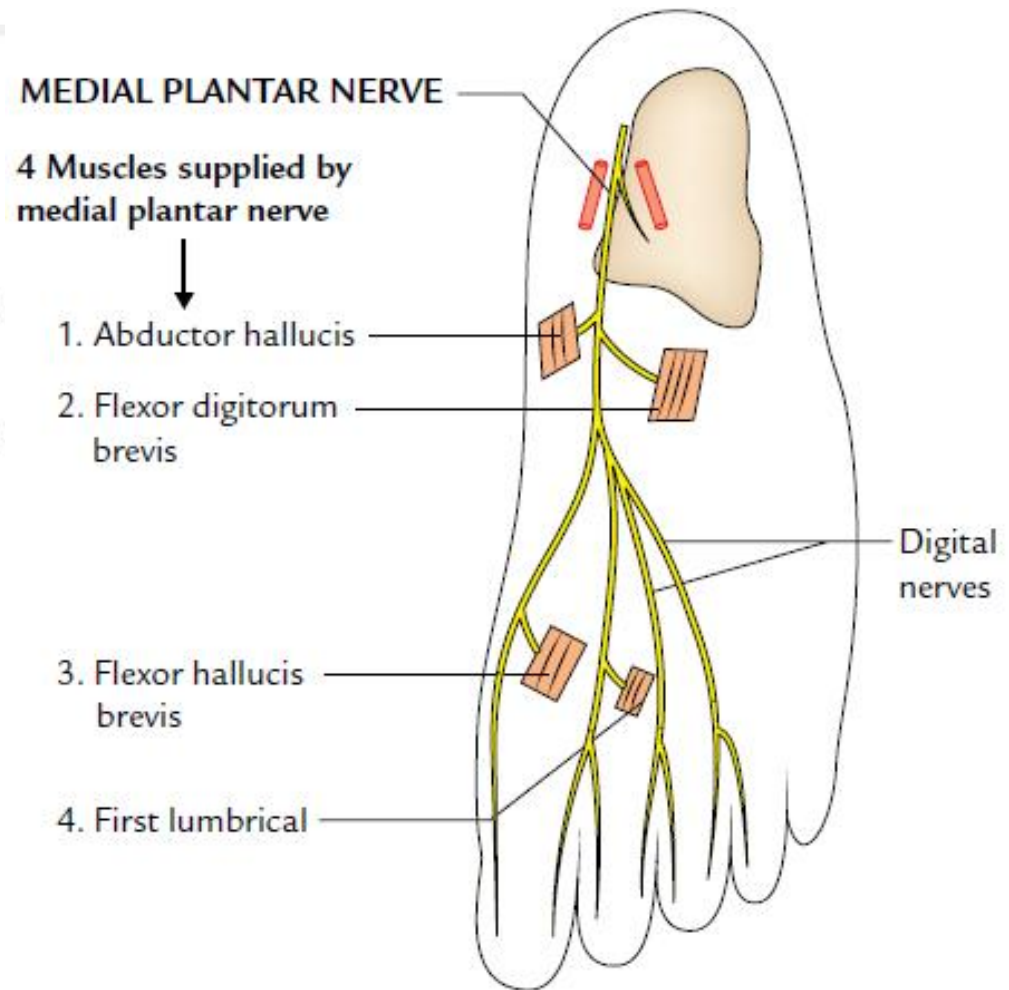
- **Cutaneous branches** to the lateral 1 and half digit
- **Motor branches to**
 - Quadratus plantae
 - Abductor digiti minimi
 - Flexor digiti minimi
 - 3 lateral lumbricals
 - 7 interossei
 - Adductor hallucis



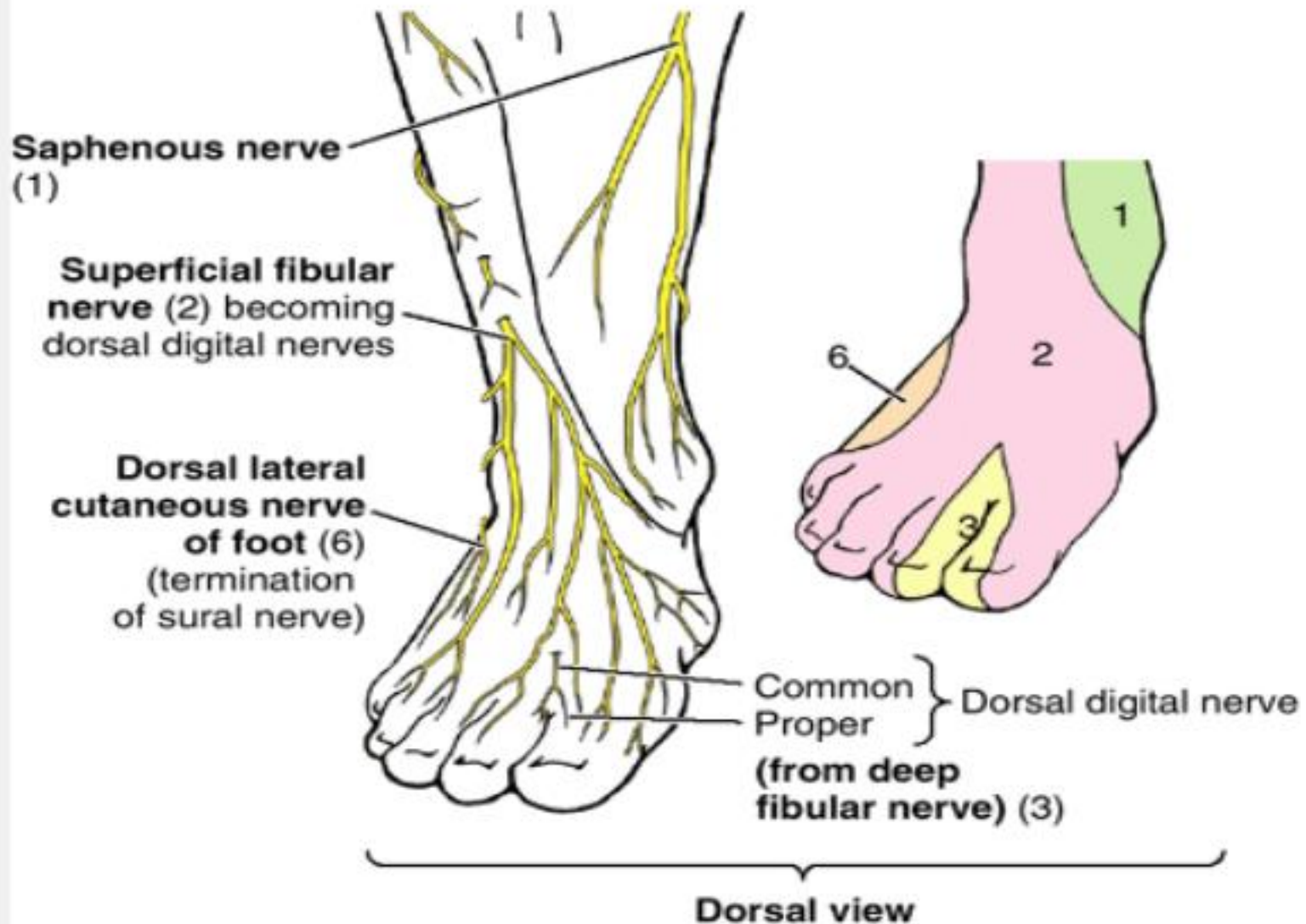
Nerves of foot

❑ Medial plantar nerve

- **Cutaneous branches** to the medial 3 and half digits
- **Motor branches to**
 - Abductor hallucis
 - Flexor digitorum brevis
 - Flexor hallucis brevis
 - Lumbrical 1



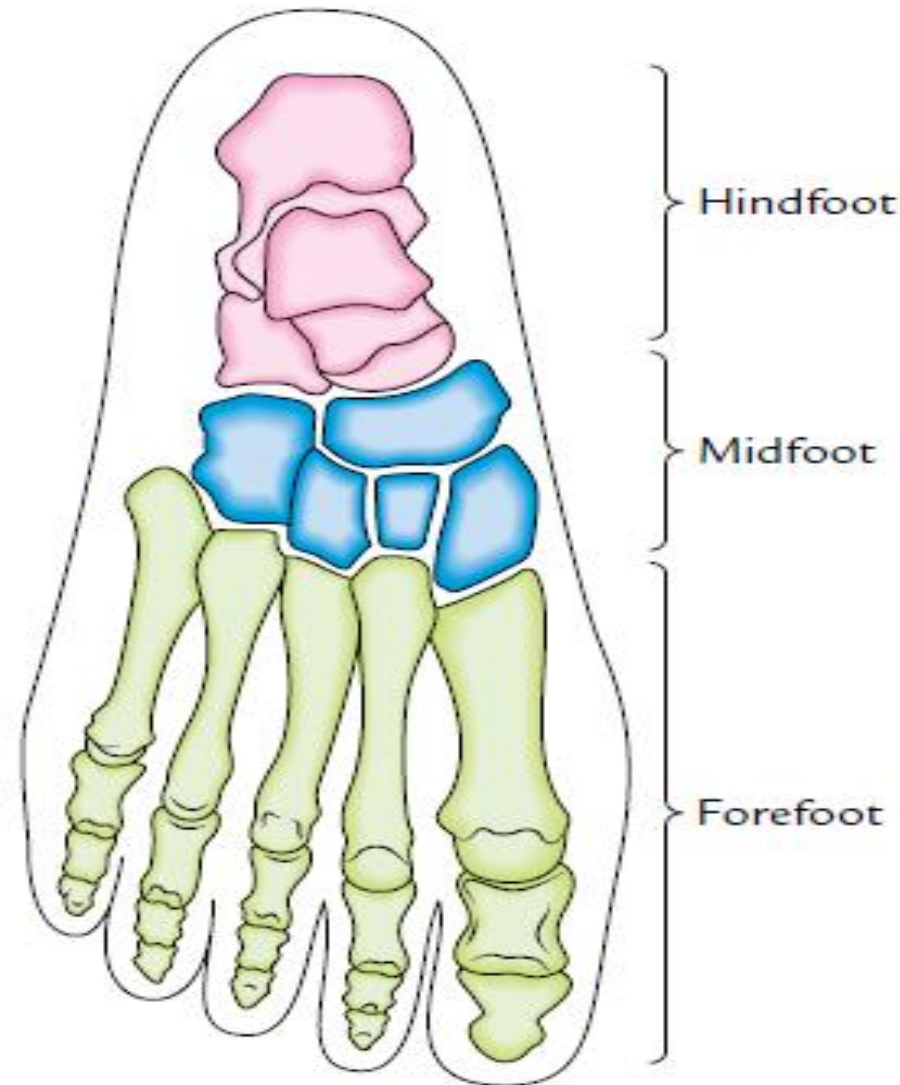
Nerves of foot



ARCHES OF THE FOOT

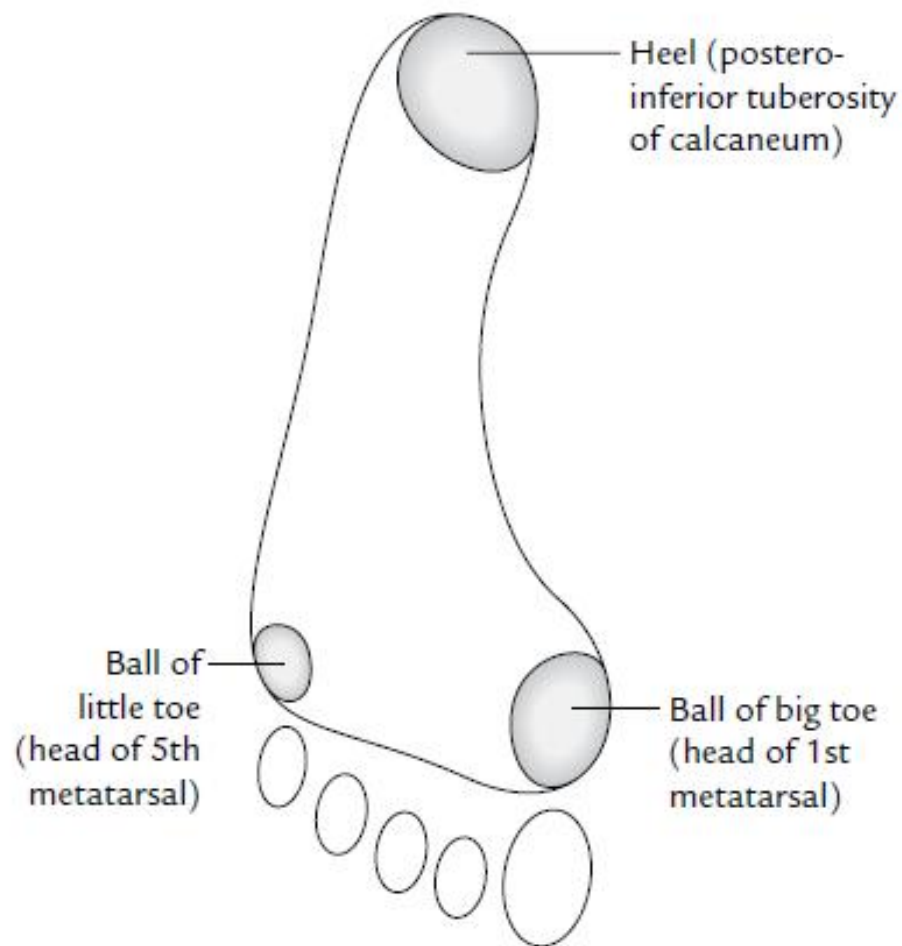
Arches of the Foot

- It is distinguishing feature of man from other primates.
- Provides platform for supporting the weight of the body during standing position.
- Provides lever to propel the body forward during running, jumping.



Arches of the Foot

- Distribution of body weight among three points
 - At posterior inferior tuberosity of calcaneum
 - At head of 1st metatarsal
 - At head of 5th metatarsal



Types of arches

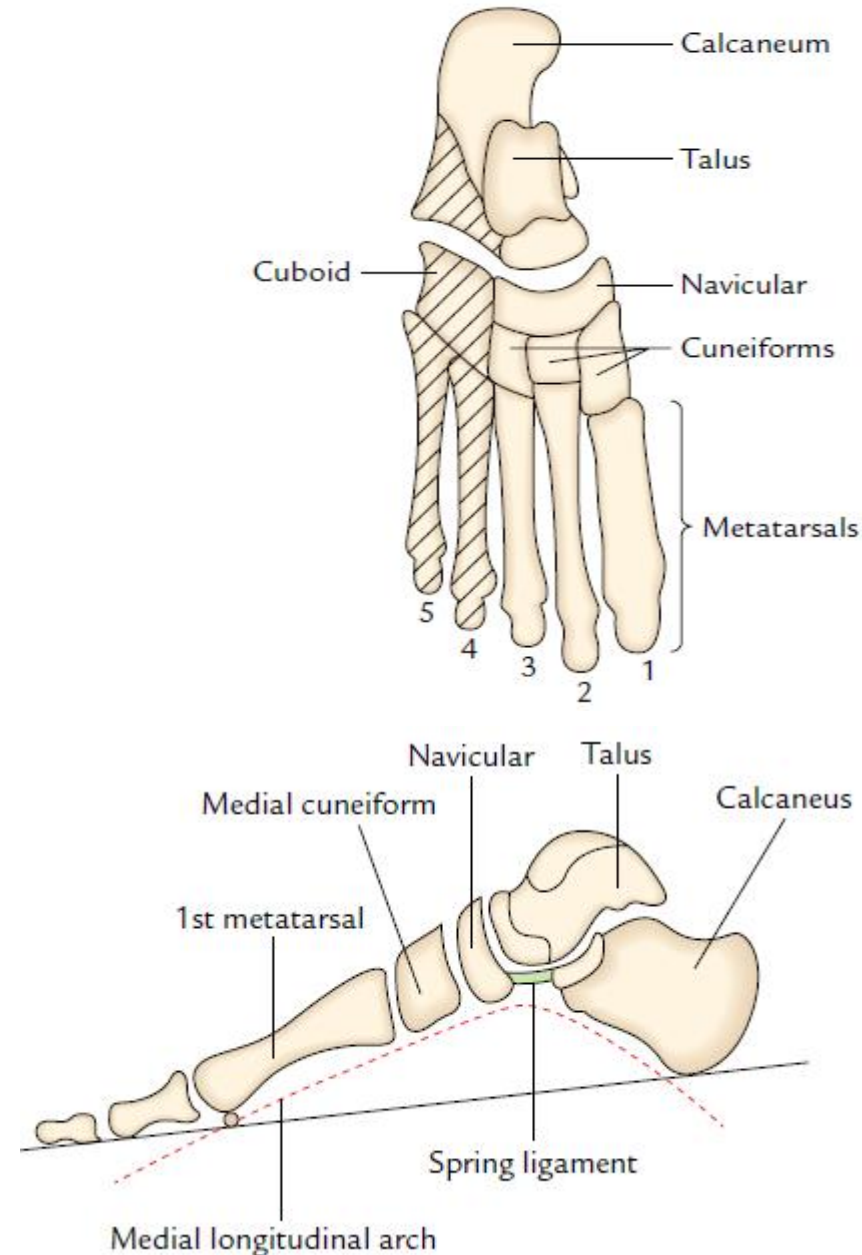
- There are two types of foot arches

1. Longitudinal foot arches

2. Transverse foot arches

- There are two **longitudinal arches** in each foot
 - Medial
 - Lateral
2. There are a series **of transverse arches** in each foot.

At the **heads of metatarsals**, the transverse arch is complete



Types of arches

1. longitudinal foot arches

Each longitudinal foot arches has pillar, summit and joints

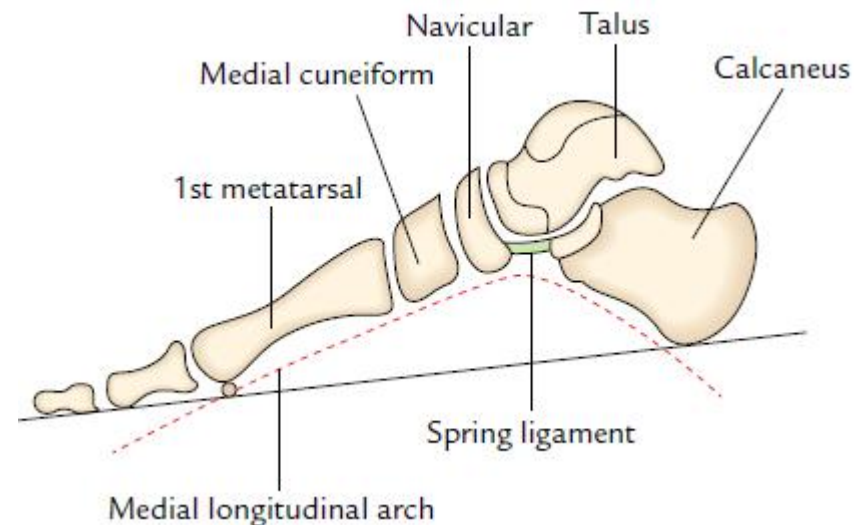
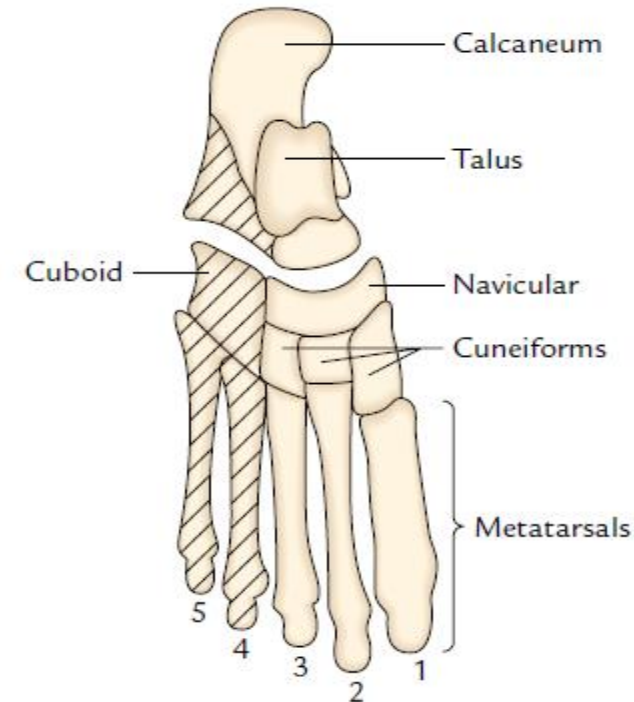
❑ Medial Longitudinal Arch

- It is formed by the **calcaneum, talus, navicular, three cuneiforms, and medial three metatarsals.**

❑ Pillars

The medial half of the calcaneum forms the **posterior pillar** of the medial longitudinal arch.

2. The heads of the medial three metatarsals form the **anterior pillar** of the medial longitudinal arch.



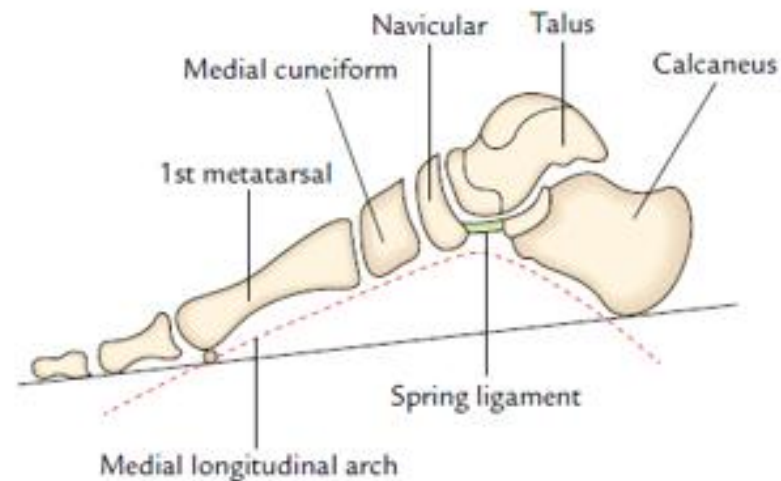
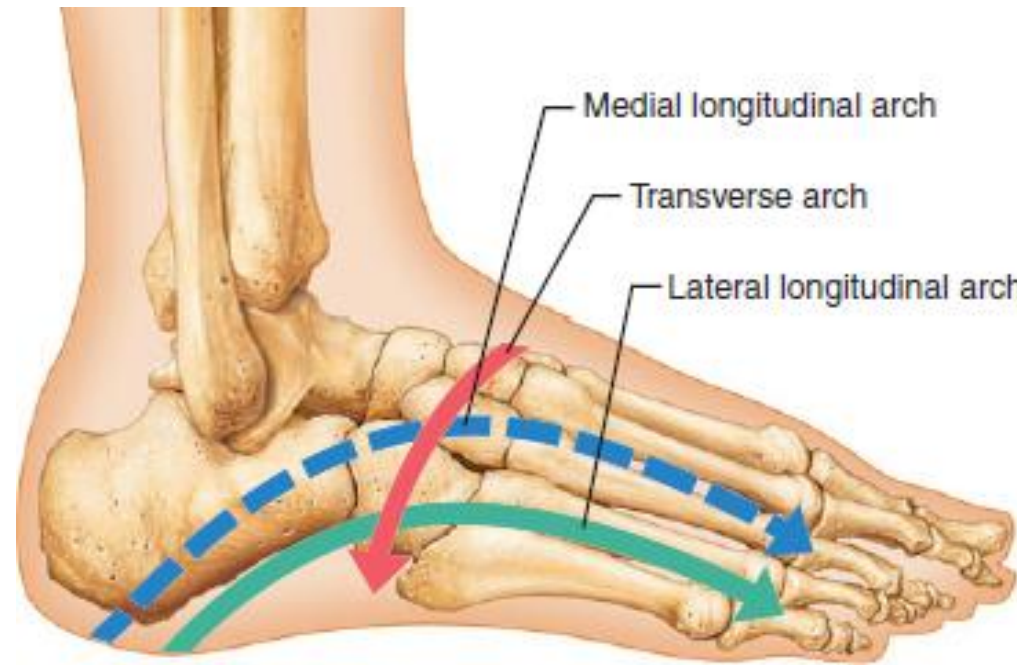
Types of arches

❑ Medial longitudinal arch's

The **talus** is the keystone of this arch.

❑ Joints

- The main joints of the **medial longitudinal arch** are
- **Talocalcaneonavicular and subtalar joints.**



Types of arches

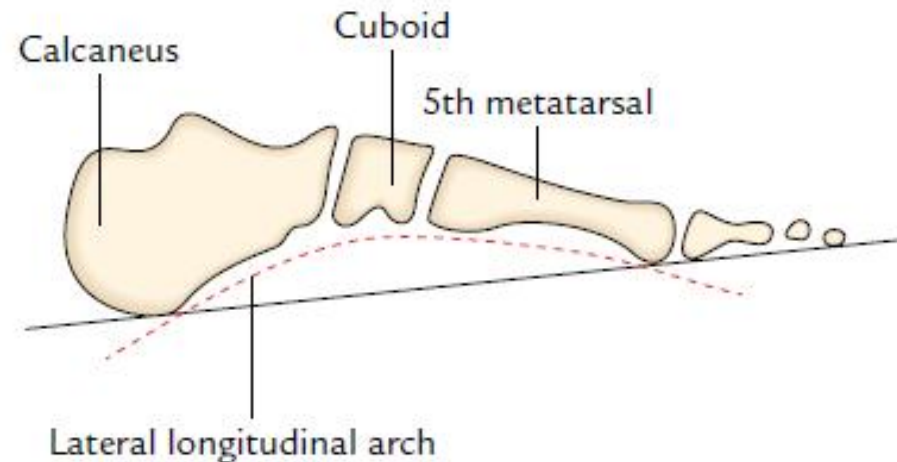
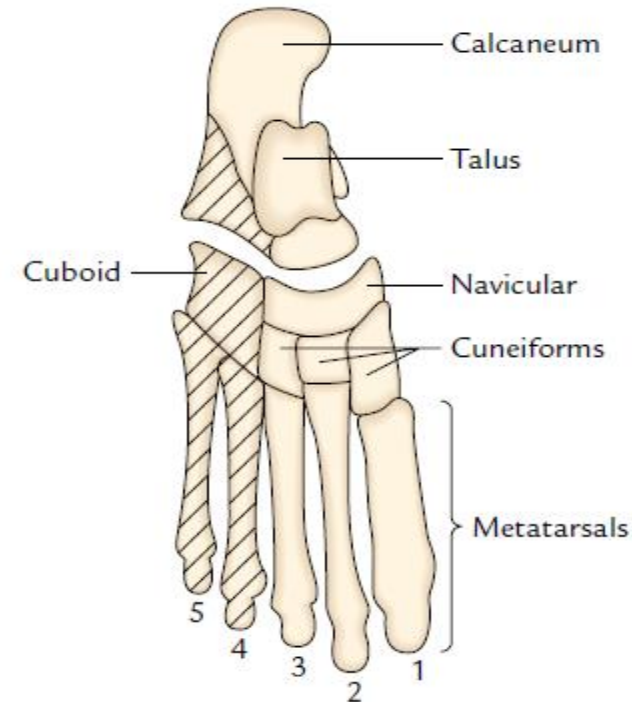
❑ Lateral Longitudinal Arch

- It is formed by the **calcaneum, cuboid, lateral two metatarsals.**

❑ Pillars

The lateral tubercle of the calcaneum forms the **posterior pillar** of the lateral longitudinal arch.

2. The heads of the lateral two metatarsals form the **anterior pillar** of the lateral longitudinal arch.



Types of arches

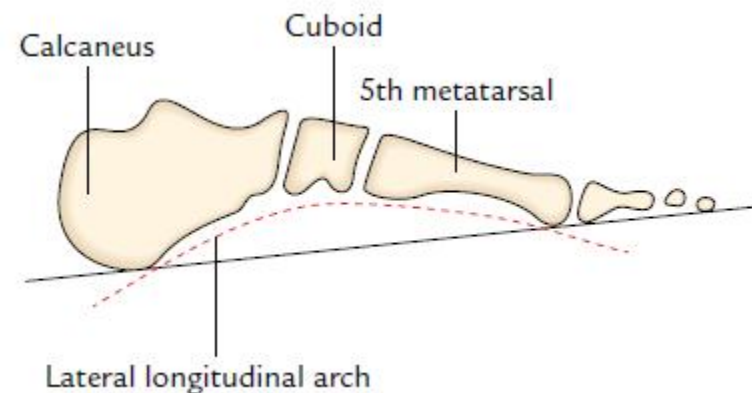
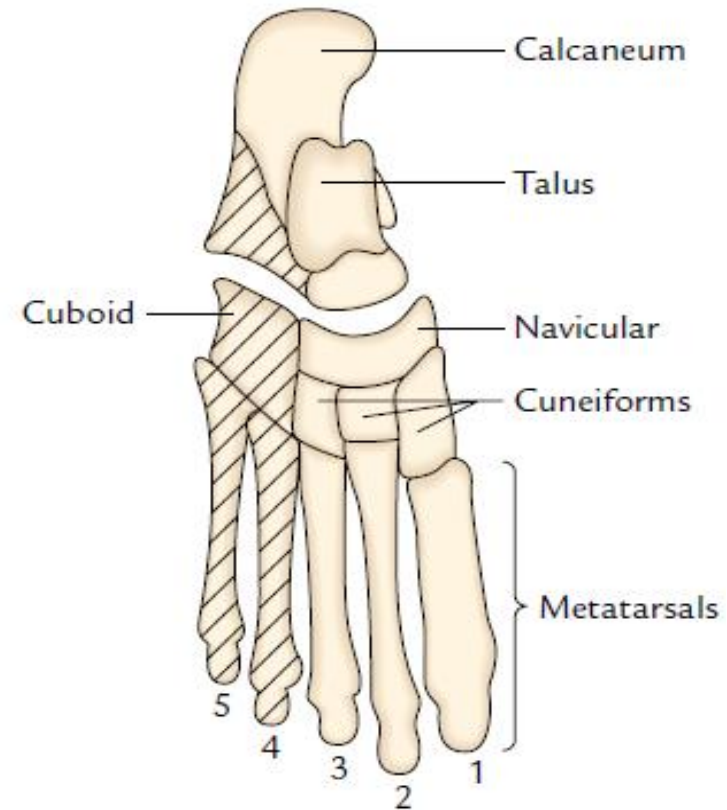
❑ Lateral longitudinal arch's...

❑ Summit

- The lateral longitudinal arch lies at the level of articular facets on the superior surface of the calcaneum (i.e., at the level of **subtalar joint**)..

❑ Joints

- The main joints of the lateral longitudinal arch are **Calcaneo-cuboid joints**.
- **NB:** The **lateral longitudinal** arch being **lower and less mobile** than the medial longitudinal arch is adapted for transmission of weight and thrusts



Types of arches

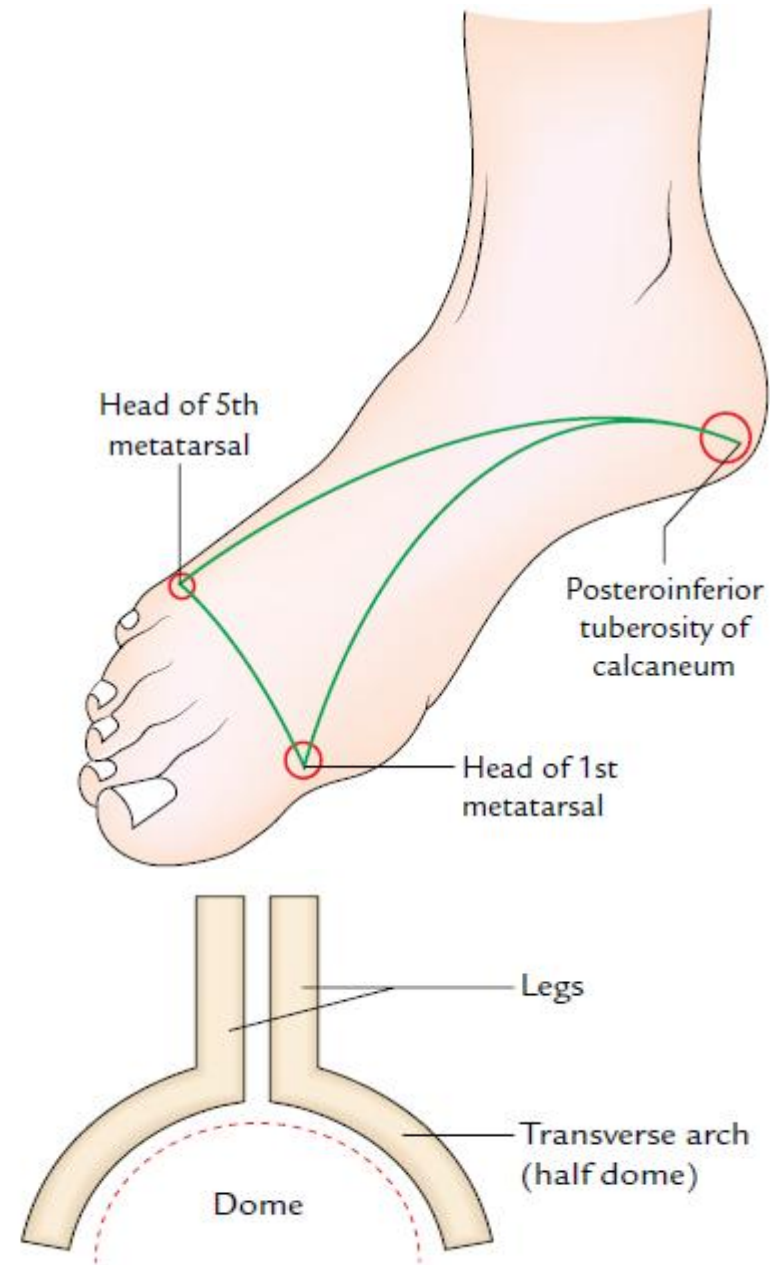
❑ Transverse arch

❑ Anterior Transverse Arch

- The heads of the metatarsals form the **anterior transverse arch**.

❑ Posterior Transverse Arch

- is formed by greater parts of the tarsus and metatarsus.

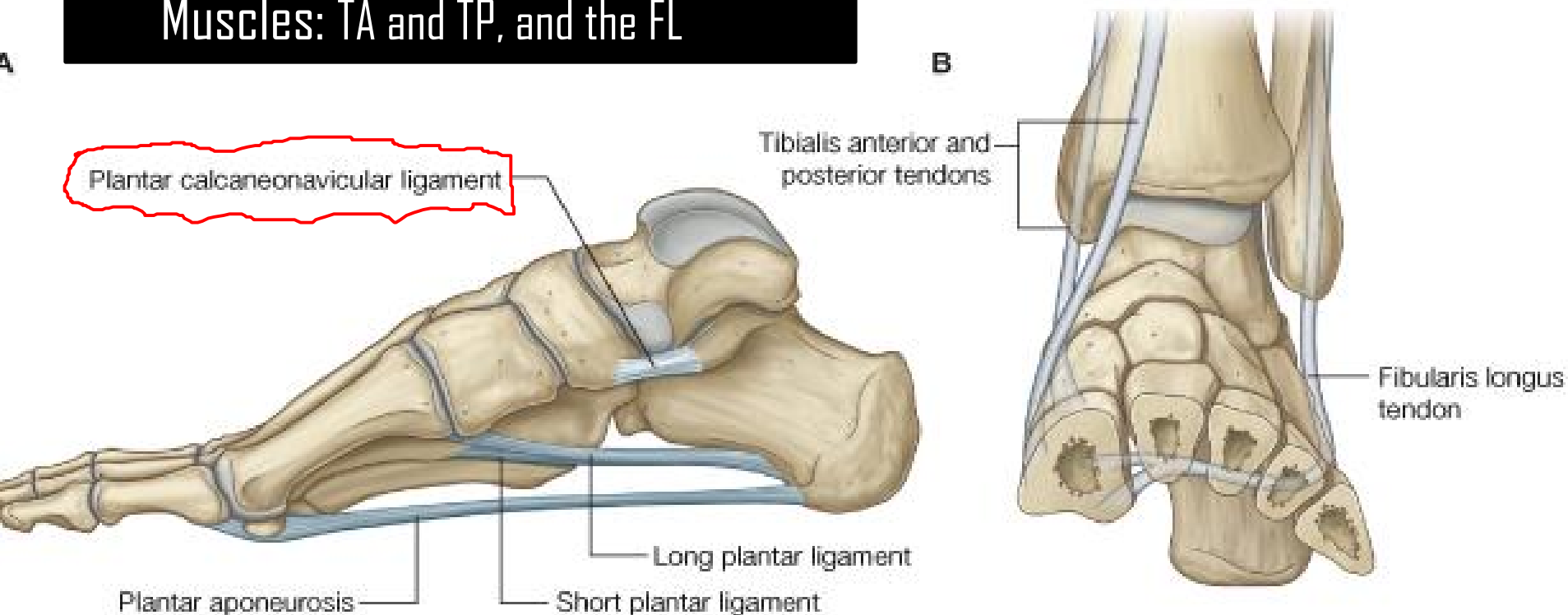


Ligament and muscle support

Ligaments:

- plantar calcaneonavicular,
- plantar calcaneocuboid, and
- long plantar ligaments, and the plantar aponeurosis

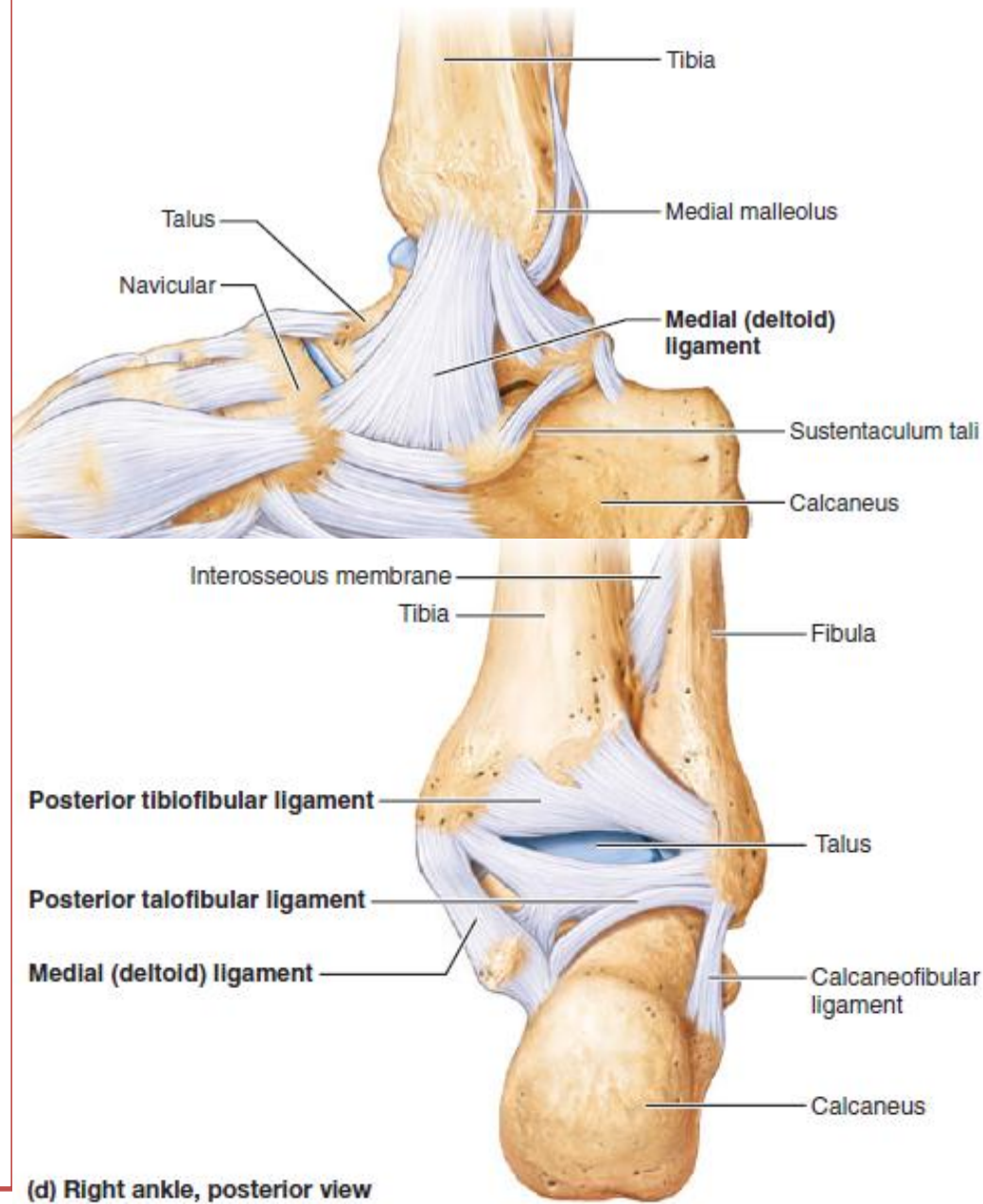
Muscles: TA and TP, and the FL



ANKLE JOINT

Ankle joint

- The joint b/n the **inferior end of tibia & fibula with talus**
- It is **hinge type synovial joint**
- It **allow dorsiflexion & plantar flexion**
- The ankle joint has capsule anteriorly & posteriorly is thin
- **Medially & laterally thickened form ligaments**
- **Medially(Deltoid)- strong**
- **Laterally –weak ligaments**



THANK YOU



@solomon T.